



《CAD与工程制图》

CAD (Computer Aided Design) and Engineering Drawing

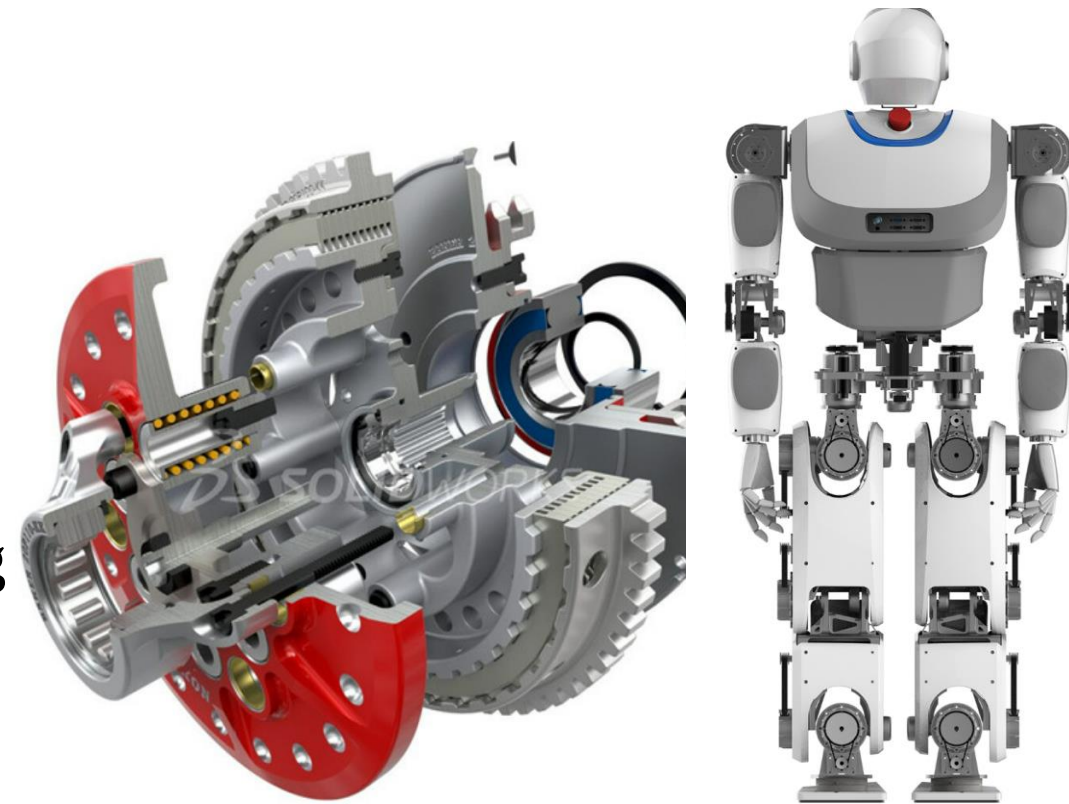
2025年 春季

Spring 2025

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Southern University of Science and Technology



Chapter II

Projection Method - National Standard - Scale Drawing

- **Projection Method**
- **National Standard**
- **Scale Drawing**

1、 Generation of Projections

Projection Phenomenon: When an object is illuminated by sunlight or artificial light, it casts a shadow on the ground or a wall.

Projection Methods: Techniques used to establish the geometric relationship between an object and its shadow.



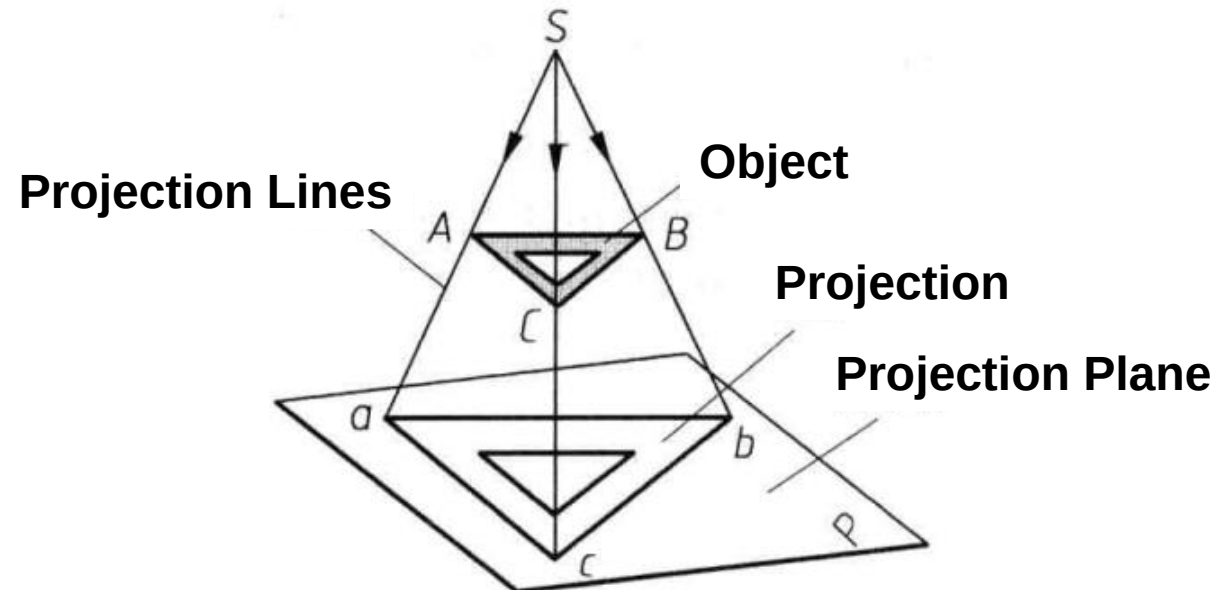
Common projection in life

1、 Generation of Projections

Projection Lines: Light rays or lines of sight.

Projection Plane: The surface on which the shadow is cast.

Three Essential Elements of Projection: Projection Lines, the Object Being Projected, and the Projection Plane — the three necessary conditions for forming a projection.



1、 Classification of Projections

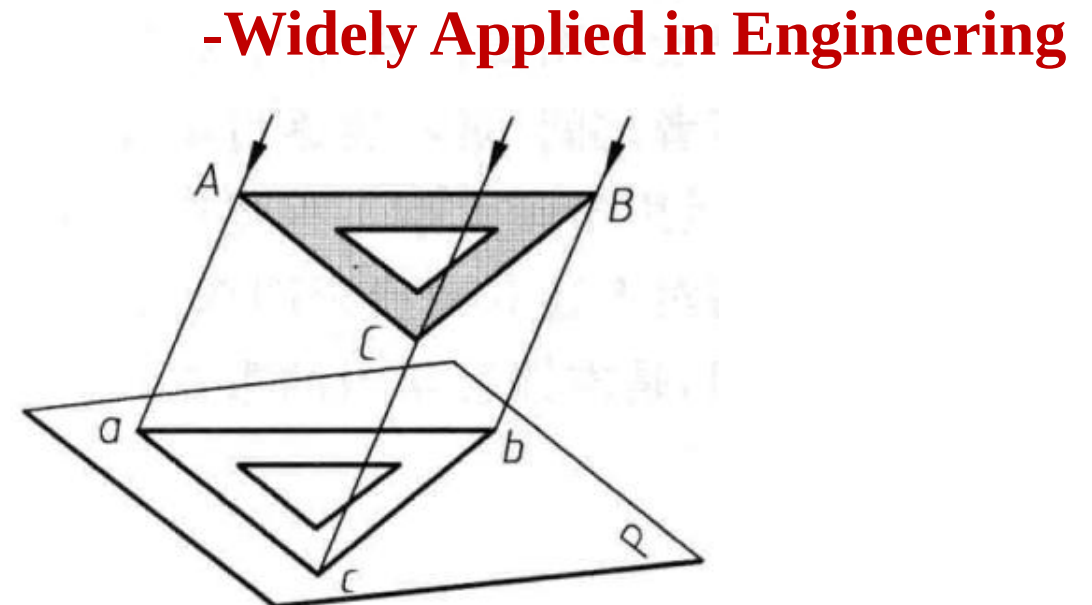
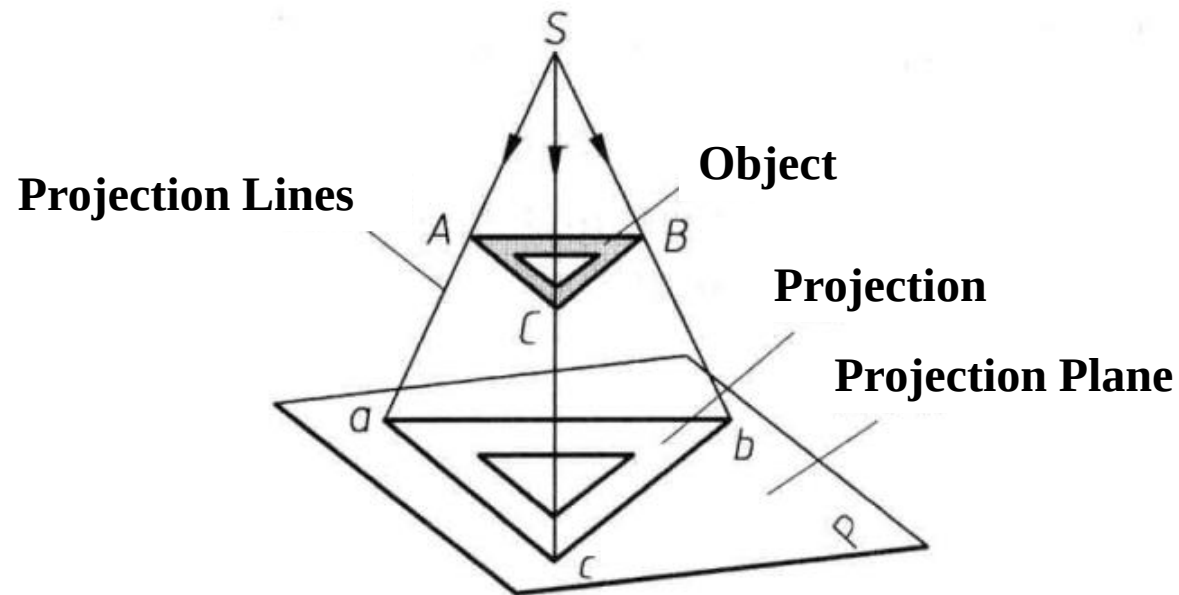
Geometric Relationship Between Projection Lines

Intersecting

Parallel

Central Projection Method

Parallel Projection Method

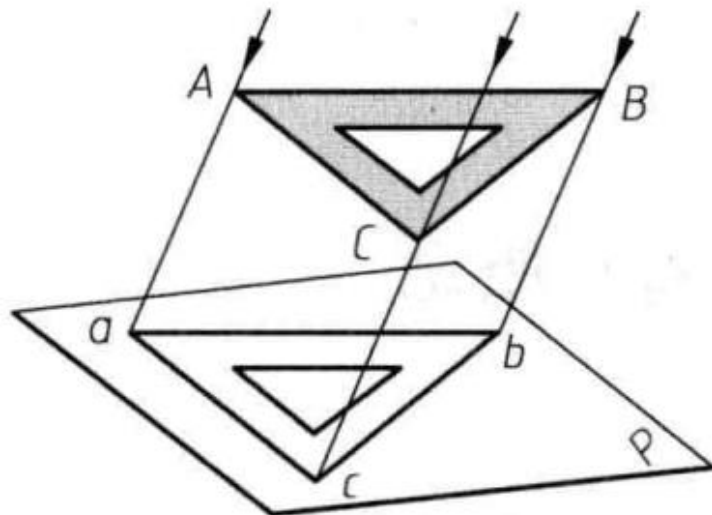


1、 Classification of Projections

Parallel Projection Lines and the Projection Plane

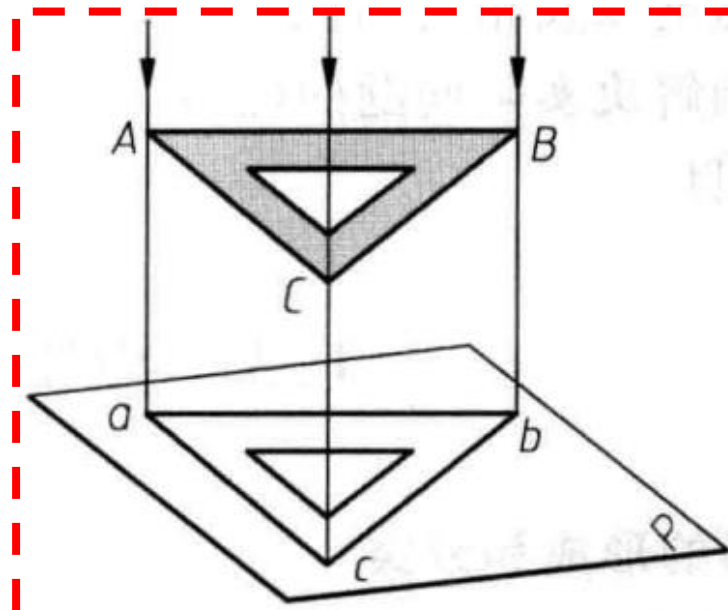
Not Perpendicular

Oblique Projection Method



Perpendicular

Orthographic Projection Method



Advantages: Accurately represents the shape of objects, convenient for drafting, good measurability, widely used in mechanical engineering, and a key focus of this course.

Disadvantages: Lack of three-dimensional perception, requires a solid understanding of engineering drawing principles and good spatial imagination skills..

- **Projection Method**
- **National Standard**
- **Scale Drawing**

ICS 35.240.10
J 04



中华人民共和国国家标准

GB/T 18229—2000

CAD 工程制图规则

Rule of CAD engineering drawing

2000-10-17 发布

2001-05-01 实施

国家质量技术监督局 发布

Standard Number:

Meaning:

GB/T

18229

2000

Standard

Standard

Year of

Code

Serial

Issue

● Codes

Number

GB——Mandatory National Standard Codes

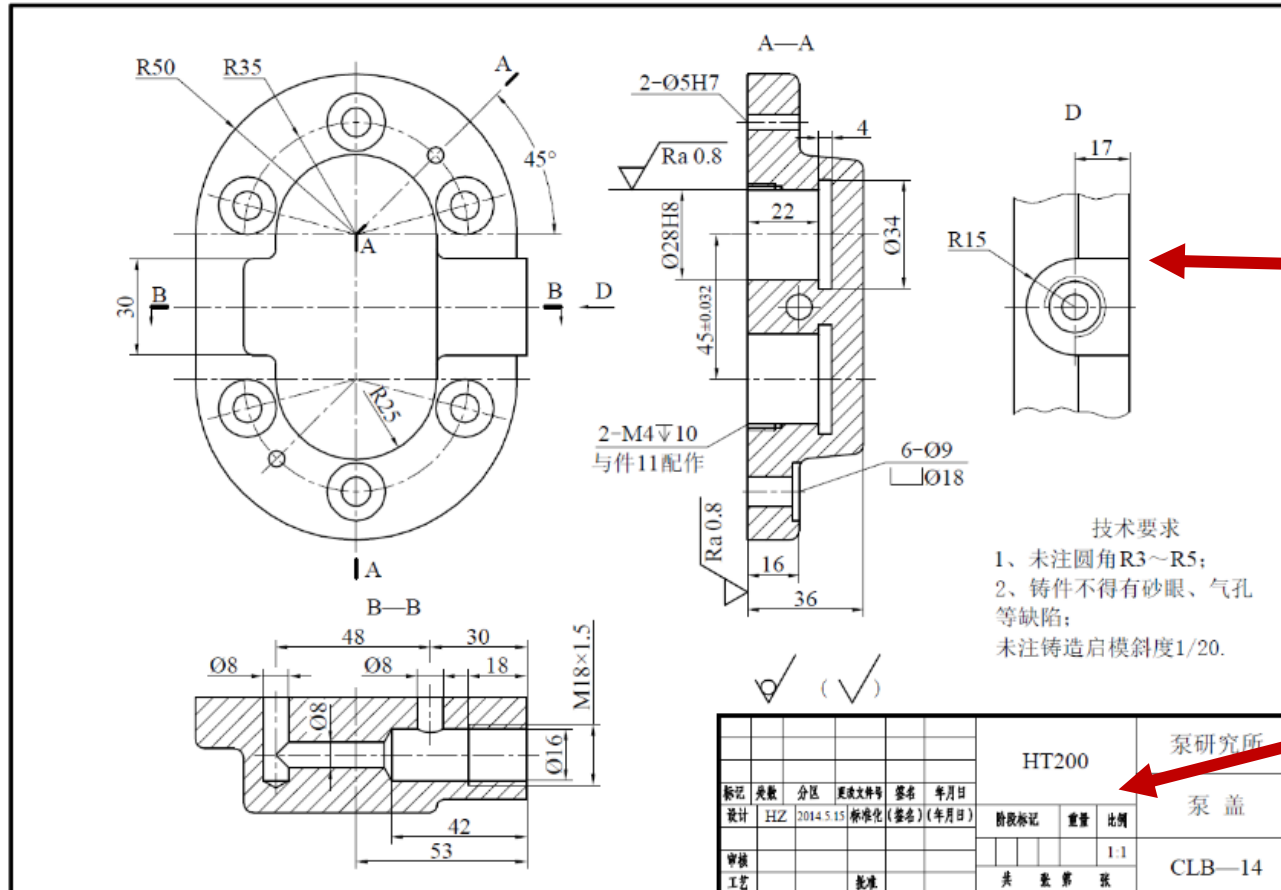
GB/T——Recommended National Standard Codes

GB/Z——National Standardization Guiding

Technical Documents, which serve as supplements to national standards.

1、 Drawing Sheet Size (GB/T 14689-2008)

Drawing Sheet Size: Refers to the dimensions of the drawing surface, composed of its width and length.



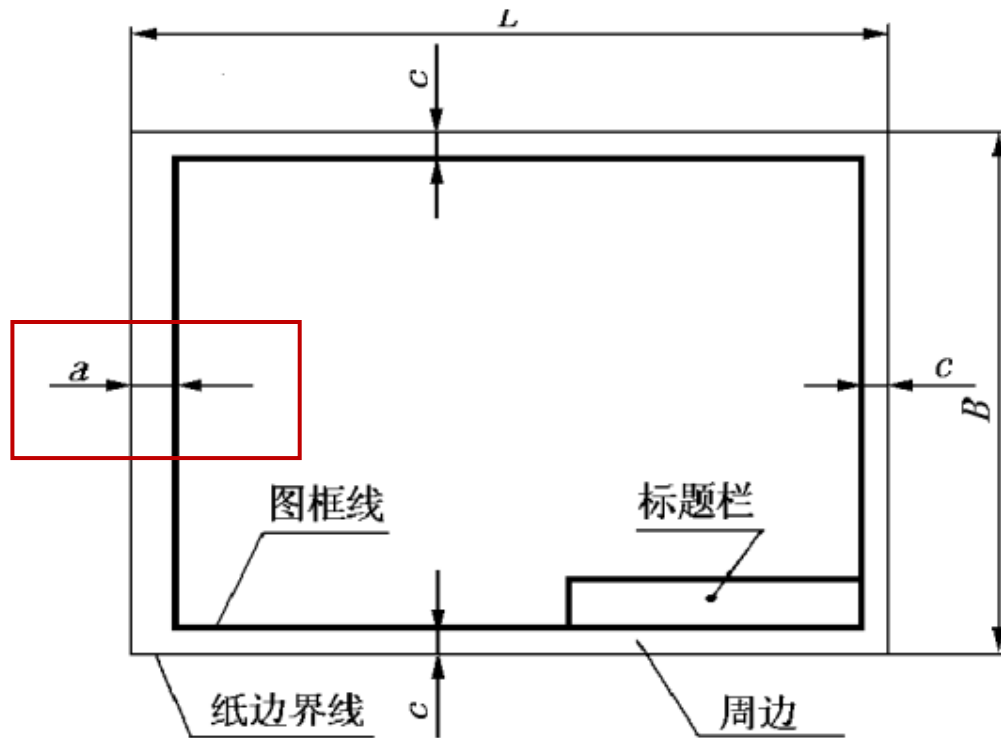
Border line

Drawing or Drawing representation

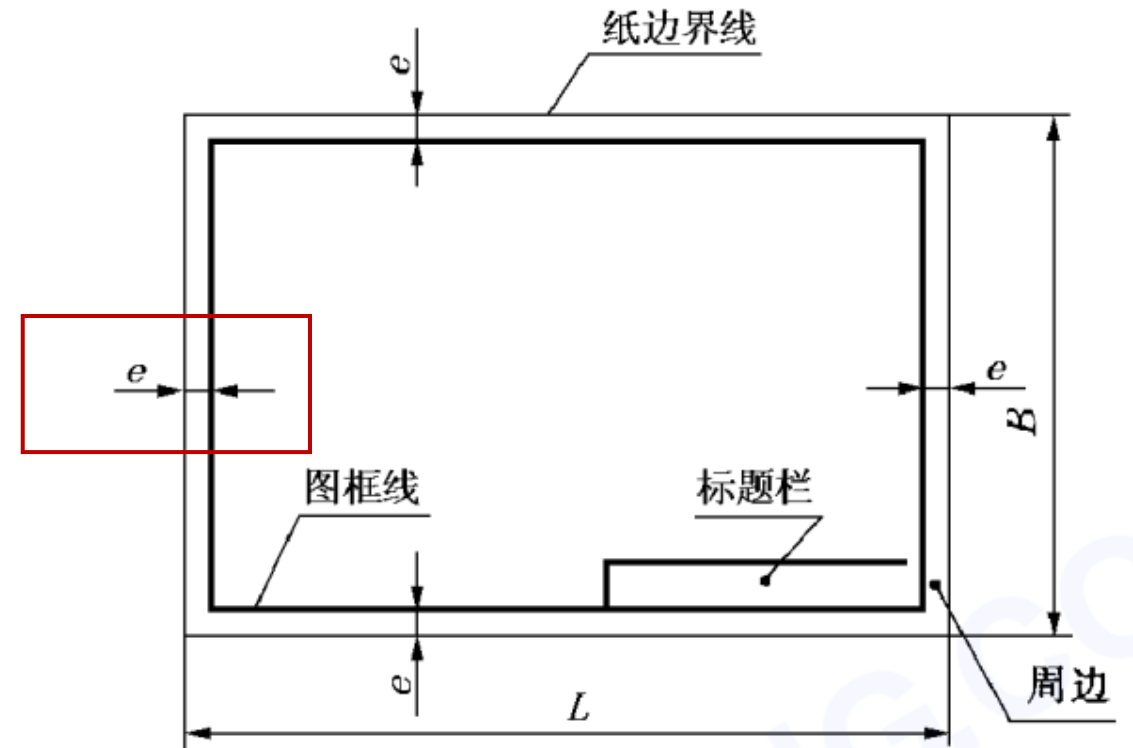
Additional symbols

Title block

1、 Drawing Sheet Size (GB/T 14689-2008)



a) 带有装订边的图纸幅面



b) 不带装订边的图纸幅面

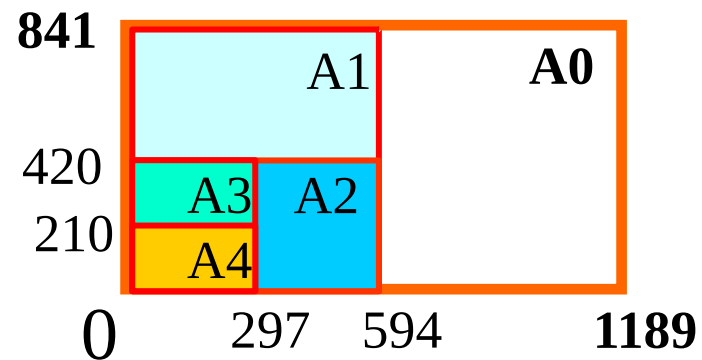
Drawing formats without binding edges are preferred.

1、 Drawing Sheet Size (GB/T 14689-2008)

Basic format dimensions (units: mm)

Format code	A0	A1	A2	A3	A4
B×L	841×1189	594×841	420×594	297×420	210×297
e	20		10		
c	10			5	
a	25				

1、 Drawing Sheet Size (GB/T 14689-2008)

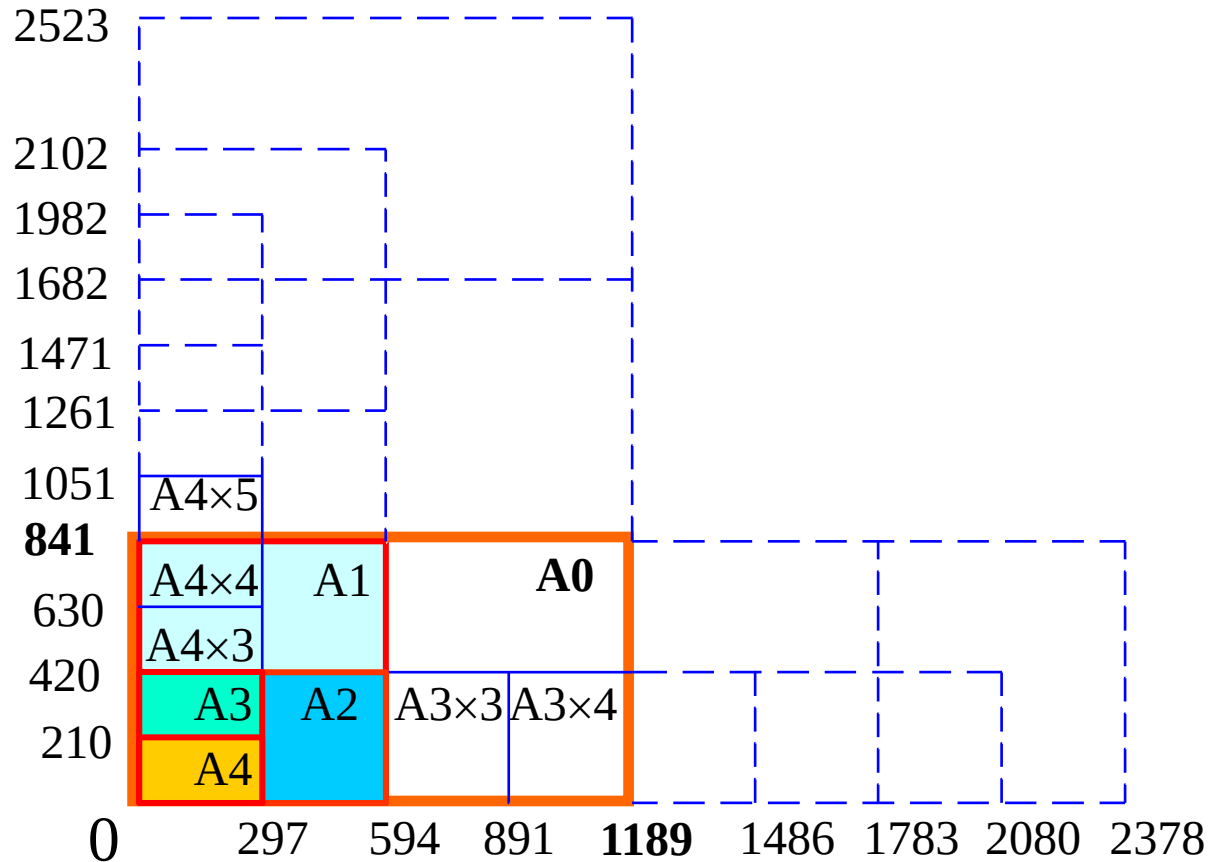


Basic format

A0: 841×1189

For other format sizes, **the drawing is folded in half along the long side to obtain a small No. 1 drawing.**

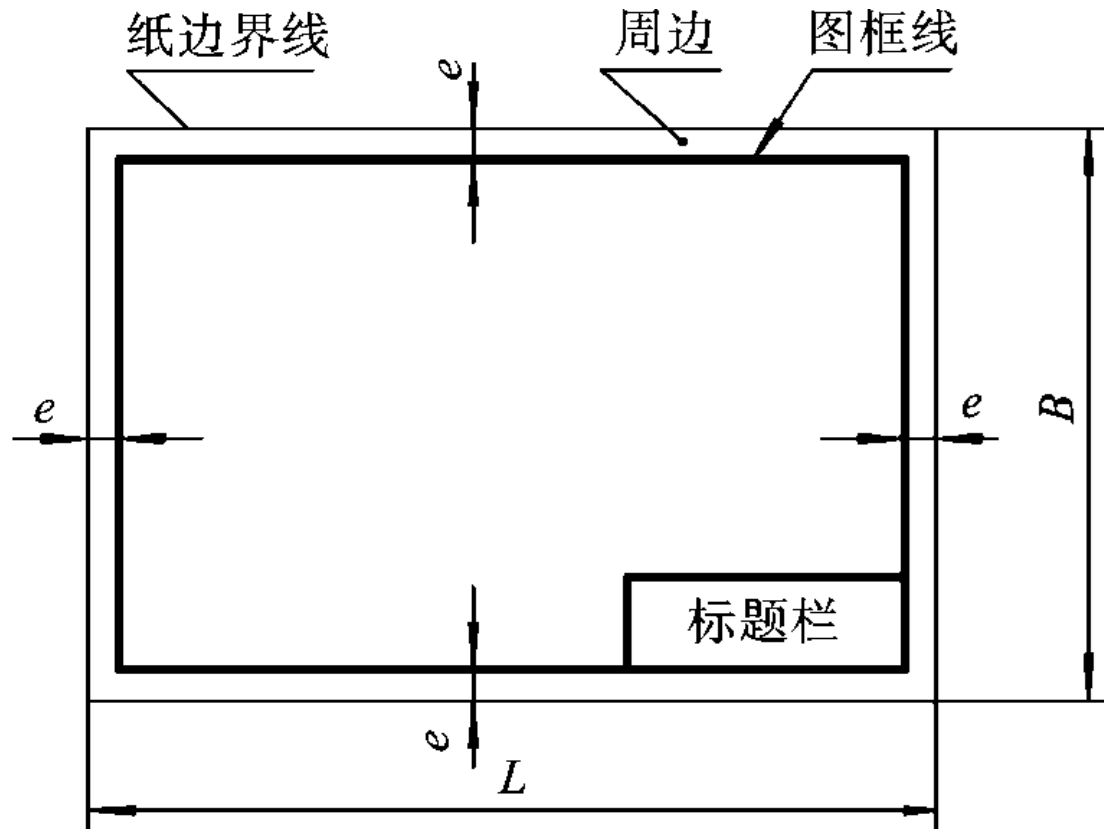
1、 Drawing Sheet Size (GB/T 14689-2008)



**Extended format drawings,
such as:
A3 ×4, commonly used in
electrical schematics.**

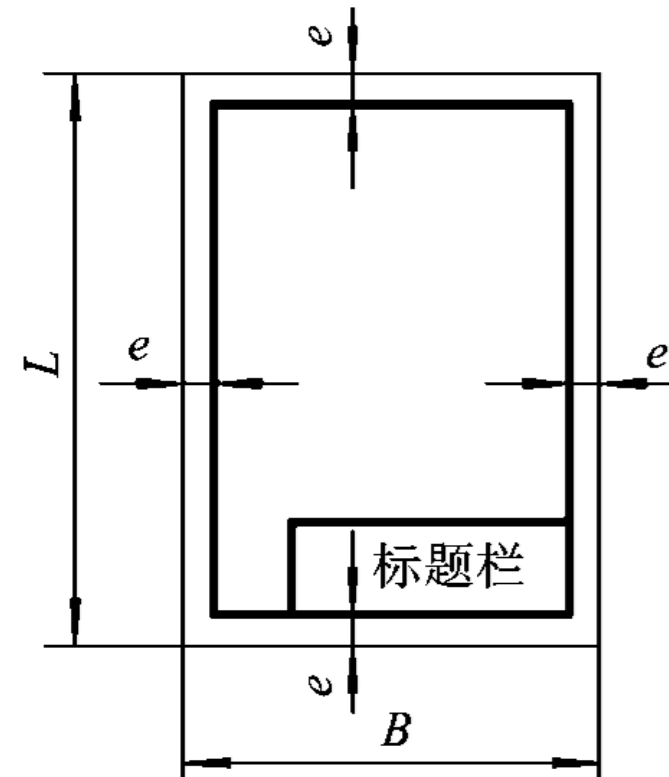
Basic format and extended format

1、Drawing format - Type (GB/T 14689-2008)



(a)

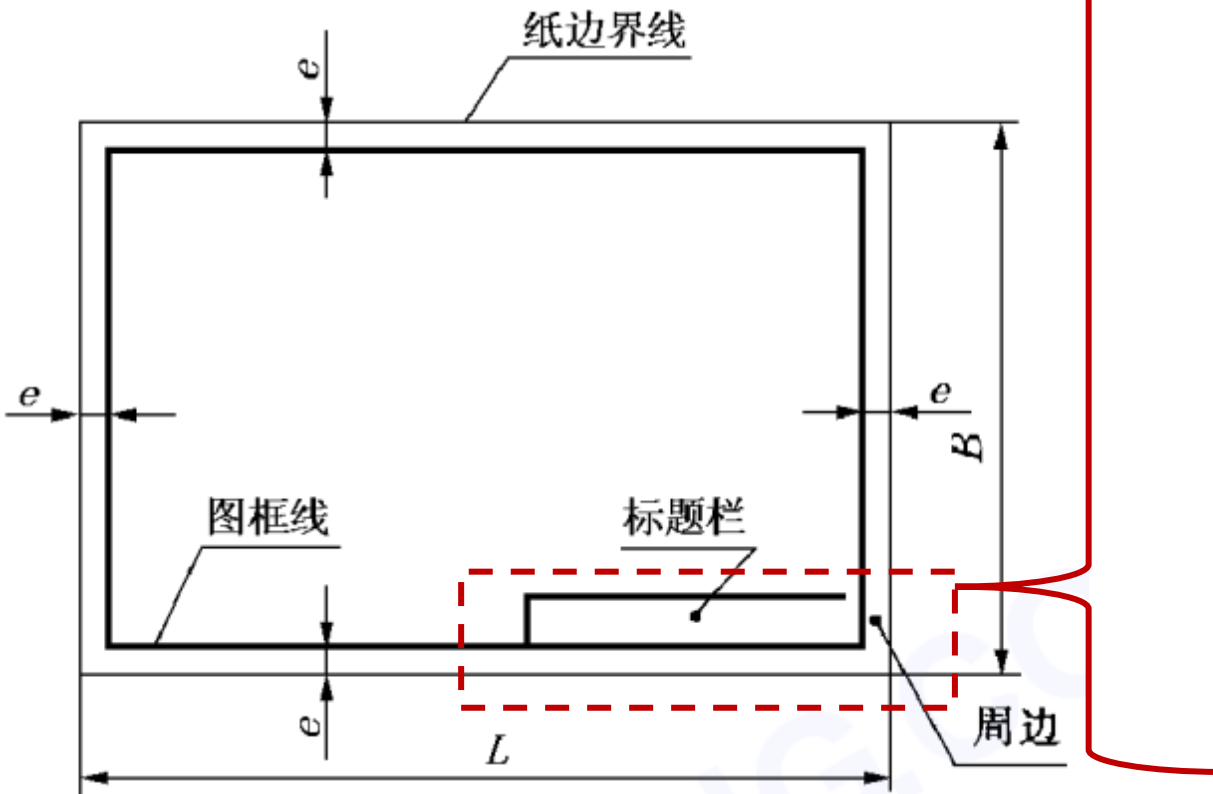
X drawing



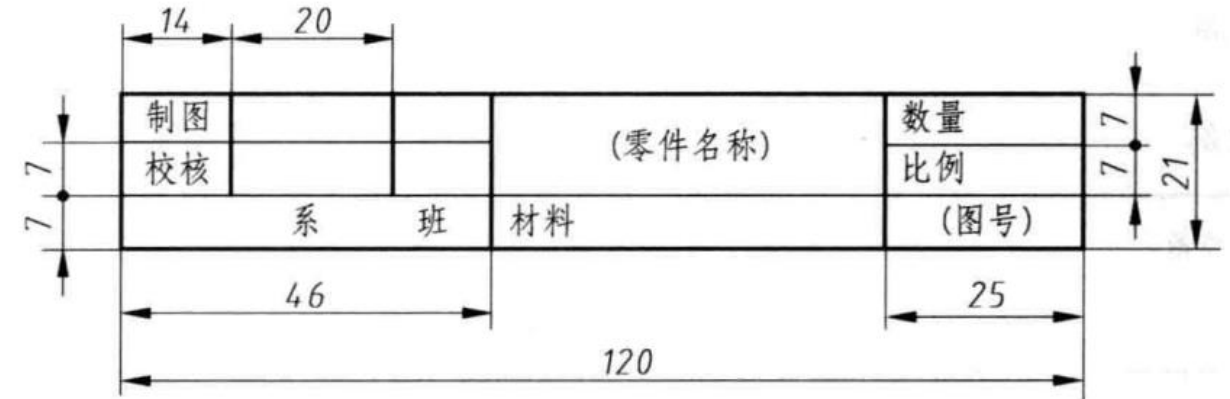
(b)

Y drawing

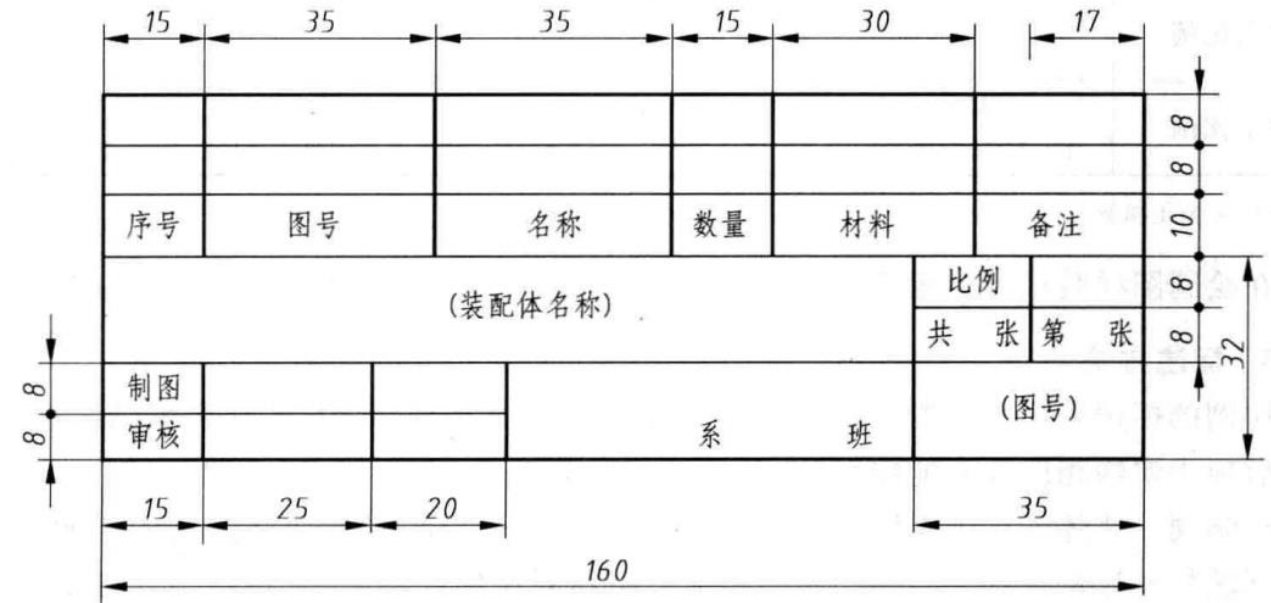
1、Drawing format - title bar (GB/T 10609.1-2008)



Title bar orientation



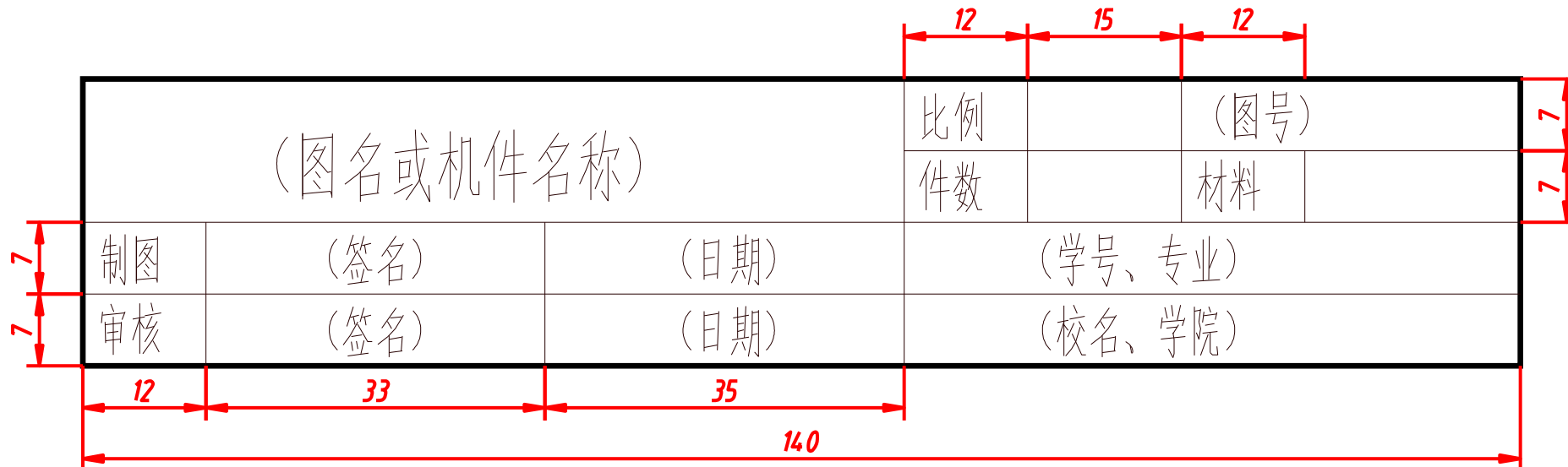
(a) Part drawing title bar



(b) Assembly drawing title bar

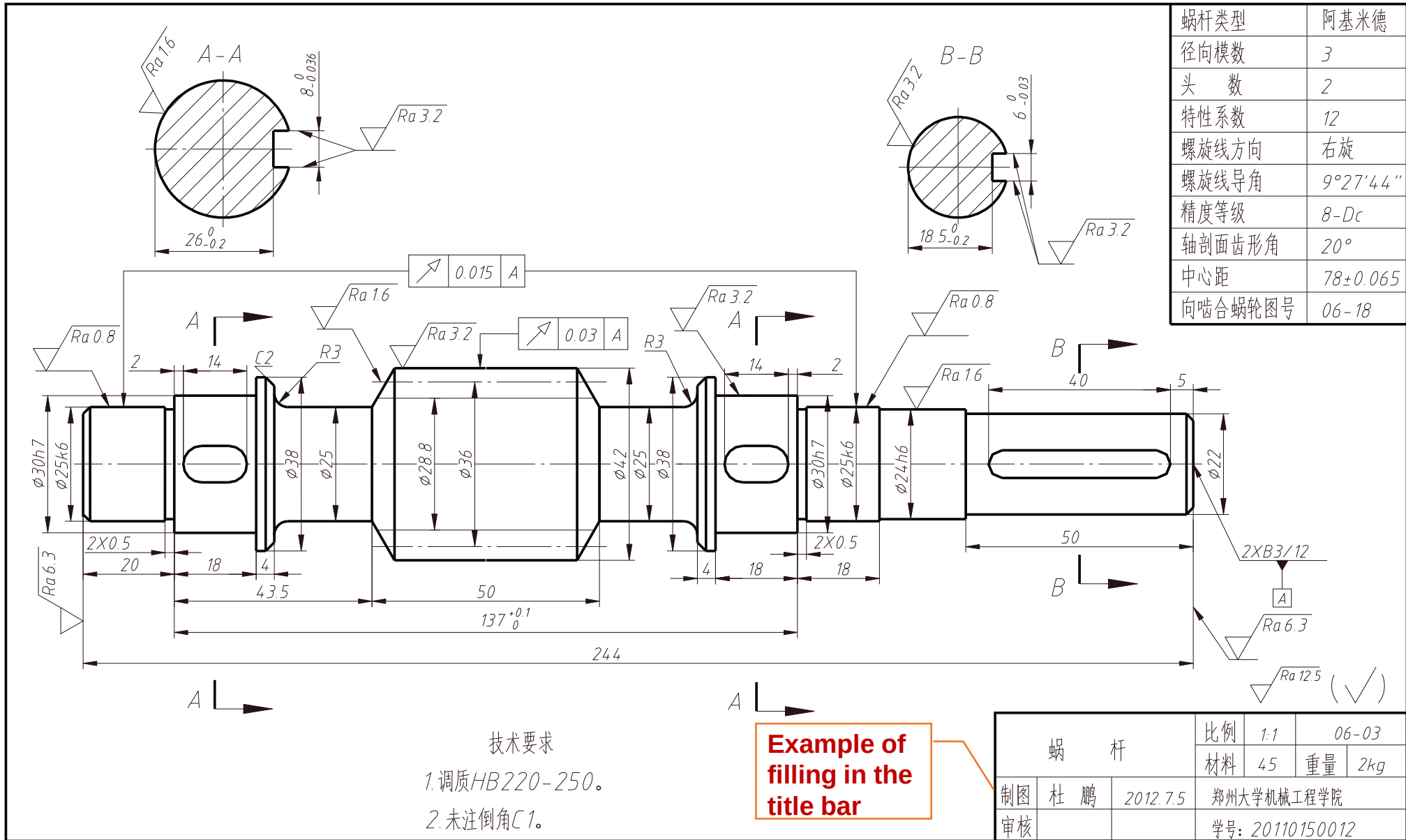
1、 Drawing format - title bar (GB/T 10609.1-2008)

In the school drawing work, in order to simplify the drawing,
the following simplification is made.

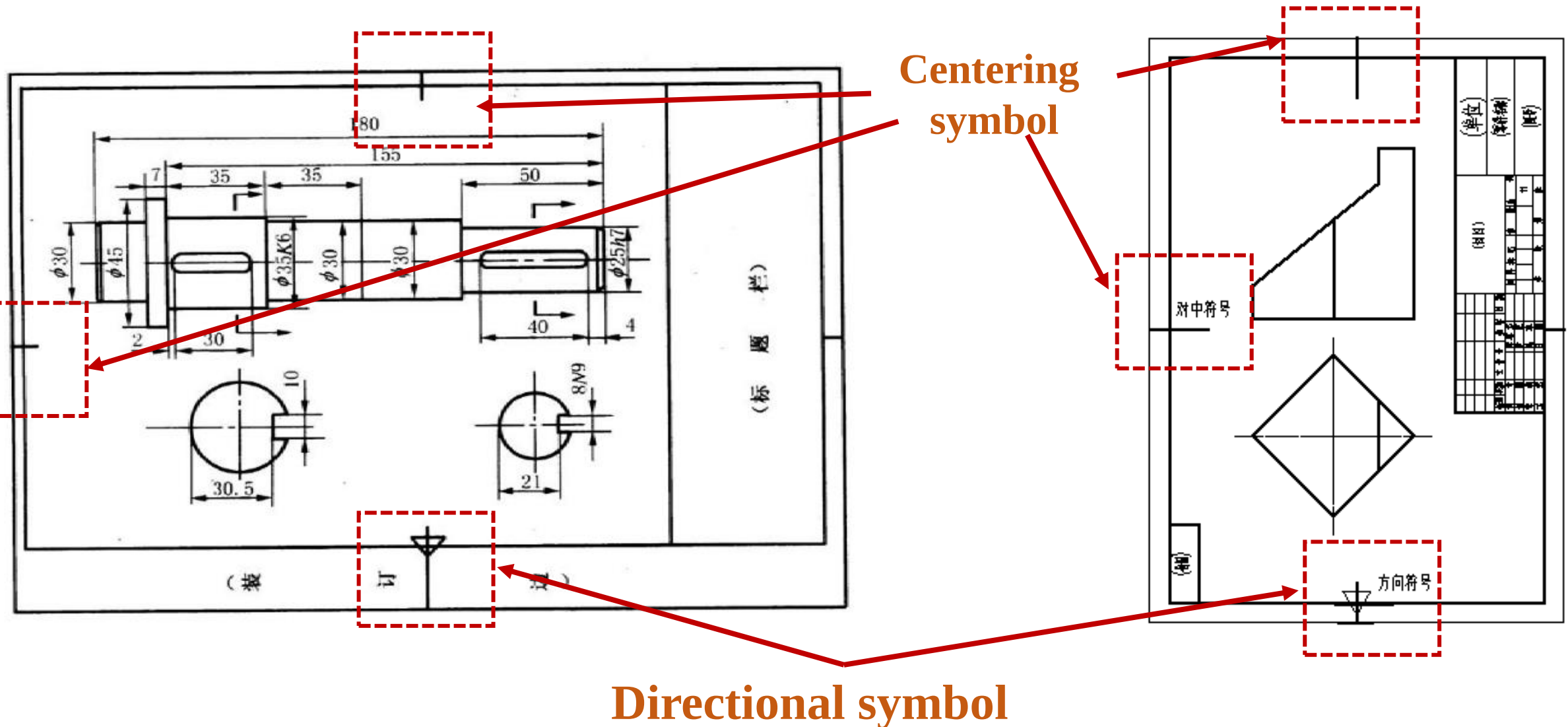


The writing direction of the text in the title bar is **the positive direction of the picture.**

National Standard - Drawing Sheet Sizes and Layouts



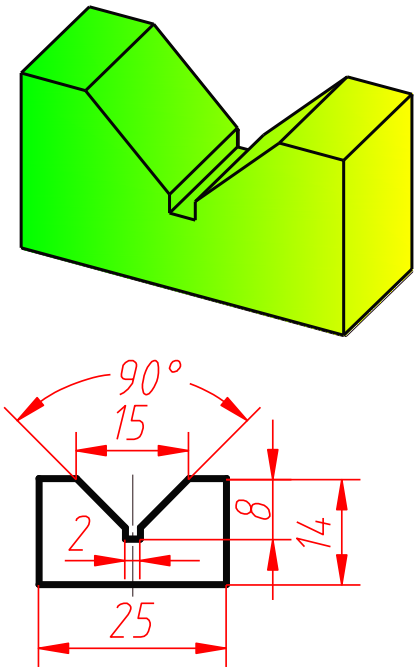
1、 Drawing format - additional symbols (GB/T 14689-2008)



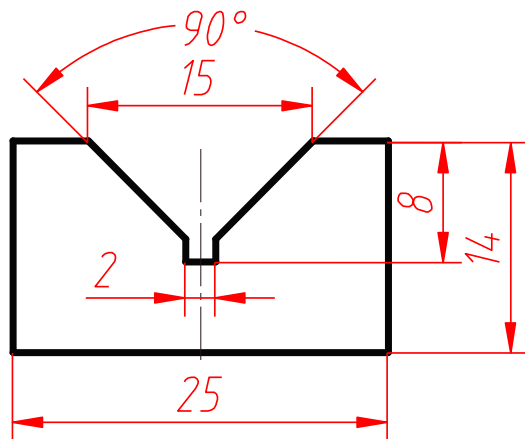
2、Drawing scale - Concept (GB/T 14690-1993)

Scale: The ratio of the linear dimensions of the figure in the drawing to the corresponding elements of the object

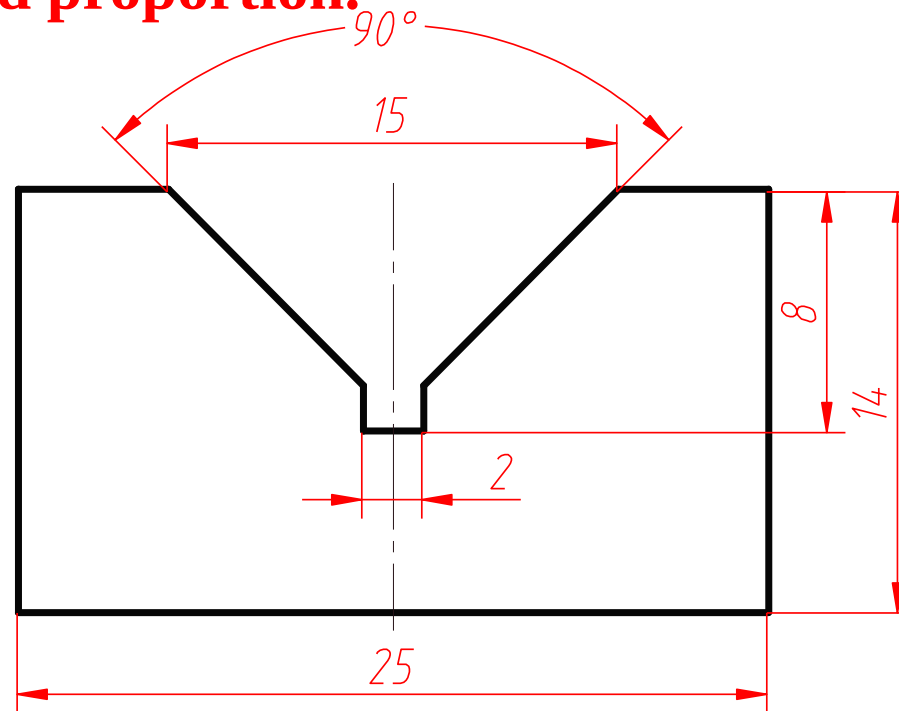
The ratio =1 is **the original proportion**, the ratio > 1 is **the enlarged proportion**, and the ratio < 1 is **the reduced proportion**.



1:2 Scale down



1:1 Original scale



2:1 Amplification ratio

Regardless of the selection of any proportion, the indicated size must be the actual size of **the physical object**.

2、Drawing scale - Select (GB/T 14690-1993)

The national standard specifies the proportion

	种类	比例				
优先采用	原值比例	1：1				
	放大比例	5:1	2:1			
		$5\times10^n:1$	$2\times10^n:1$	$1\times10^n:1$		
	缩小比例	1:2	1:5	1:10		
		$1:2\times10^n$	$1:5\times10^n$	$1:1\times10^n$		
必要时采用	放大比例	4:1	2.5:1			
		$4\times10^n:1$	$2.5\times10^n:1$			
	缩小比例	1:1.5	1:2.5	1:3	1:4	1:6
		$1:1.5\times10^n$	$1:2.5\times10^n$	$1:3\times10^n$	$1:4\times10^n$	$1:6\times10^n$

Drawing: Real = 1:1 (real)

Pattern: Real = 2:1 (enlarge)

Drawing: Physical = 1:2 (reduced)

注：n 为正整数。

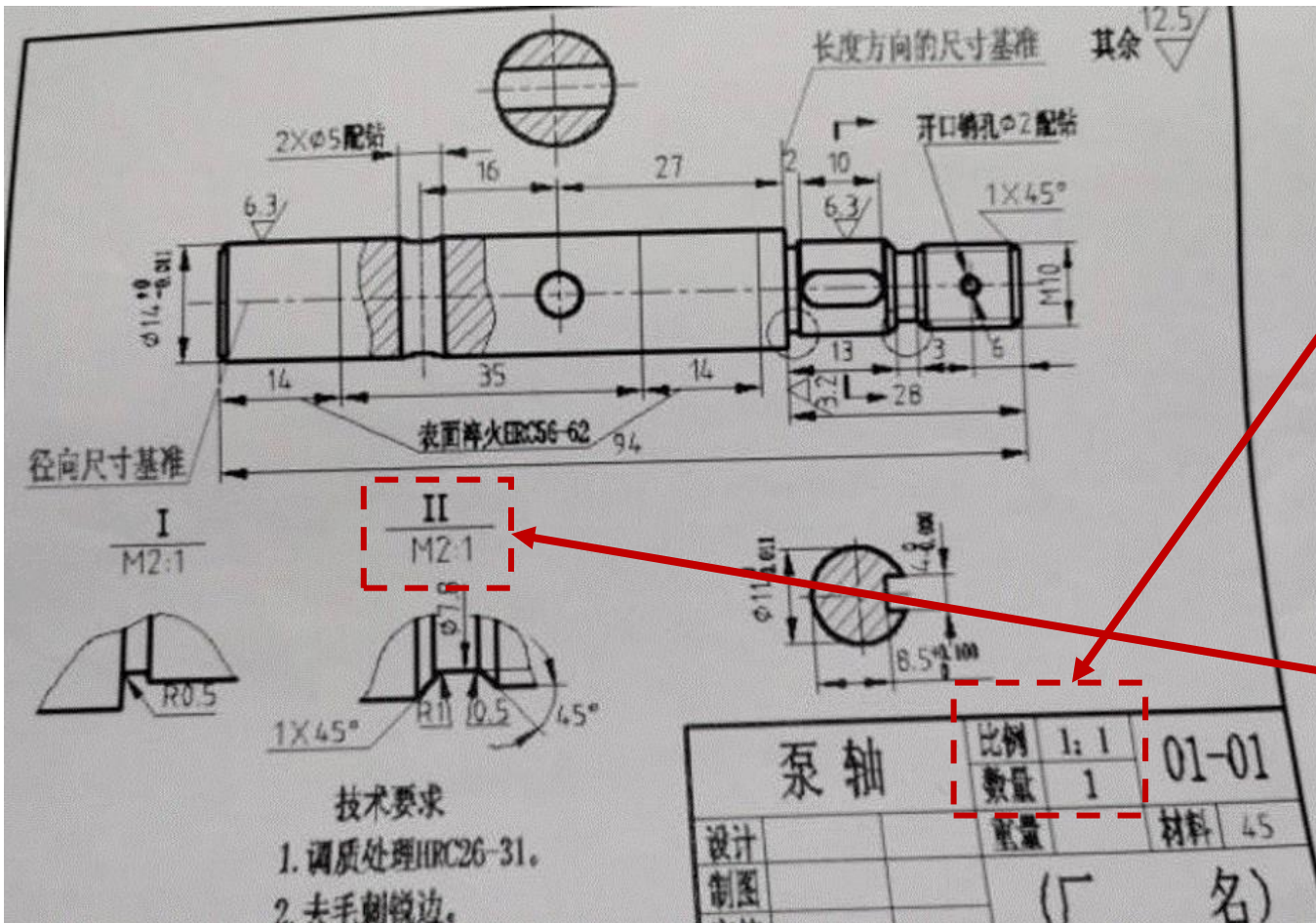
Draw as close as possible to the real size of the object to facilitate reading and spatial thinking

2、Pattern scale - Position (GB/T 14690-1993)

It is usually indicated in the scale column of the title bar.

$$\frac{I}{2:1} \quad \frac{A}{1:100} \quad \frac{B-B}{2.5:1}$$

For local views, it can be written **above the view name.**



2、 Pattern line - Type of pattern line (GB/T 17450-1998)

Common drawing line

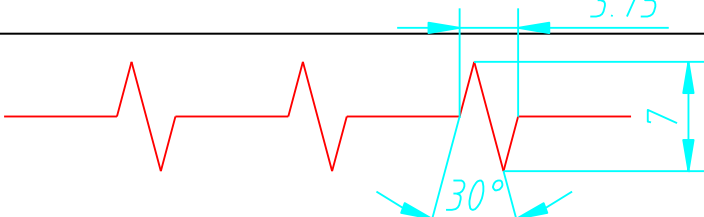
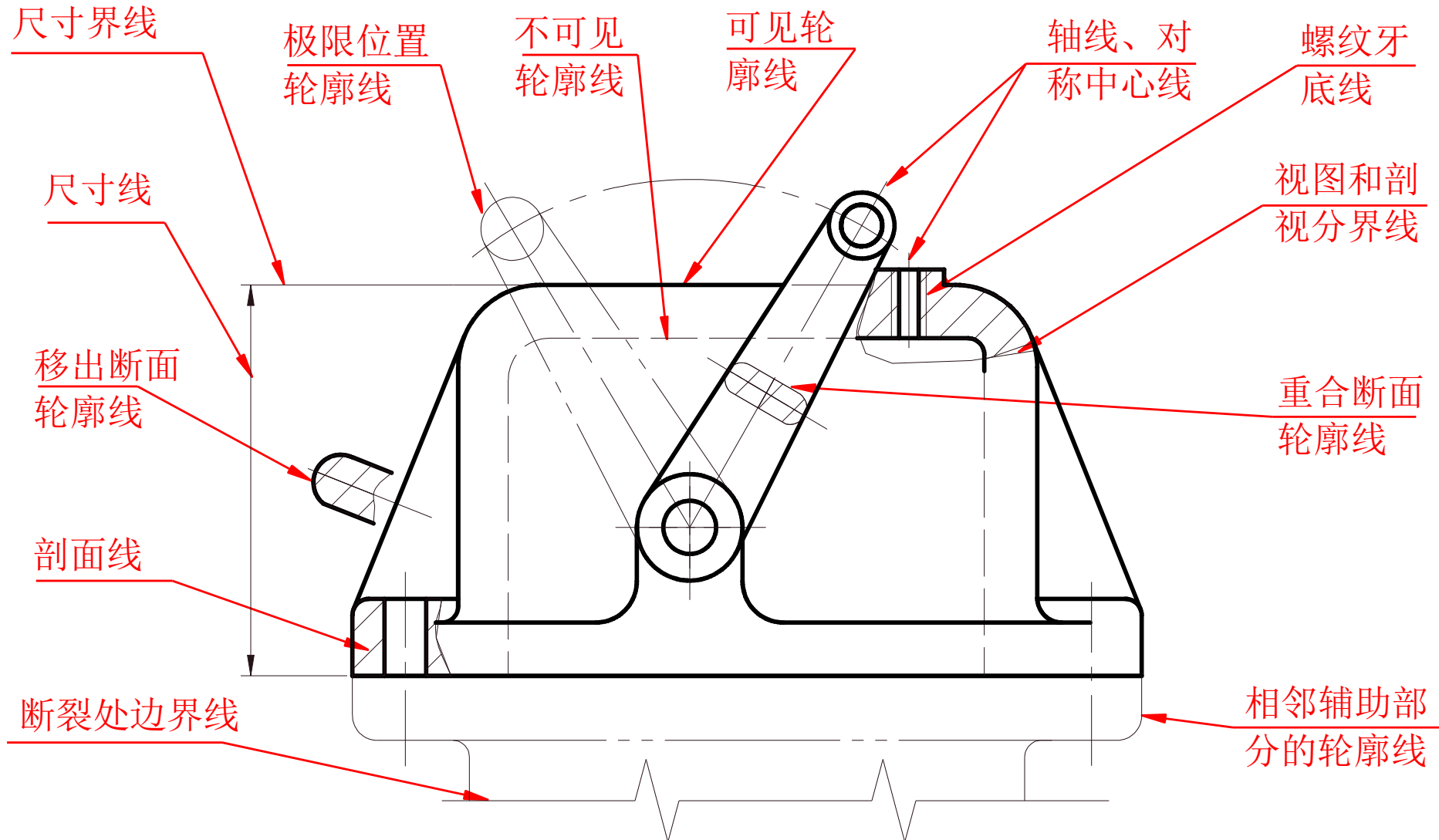
Lines	Line type	DrawWidth	Example
Heavy line		d	可见轮廓线 可见棱边线
Fine dotted line		$0.5d$	不可见轮廓线 不可见棱边线
Fine line		$0.5d$	尺寸线及尺寸界线 剖面线、引出线、重合断面的轮廓线 螺纹的牙底线及齿轮的齿根线 分界线及范围
Break line		$0.5d$	断裂处的边界线 视图和剖视的分界线
Draw lines in fine dots		$0.5d$	轴线、对称中心线 轨迹线、节圆及节线
双折线		$0.5d$	断裂处的边界线 视图和剖视的分界线

Diagram application legend



Line drawing is an example

- Judge the correctness of the following two figures

The dotted line is too short, should exceed the outline line 2-5mm

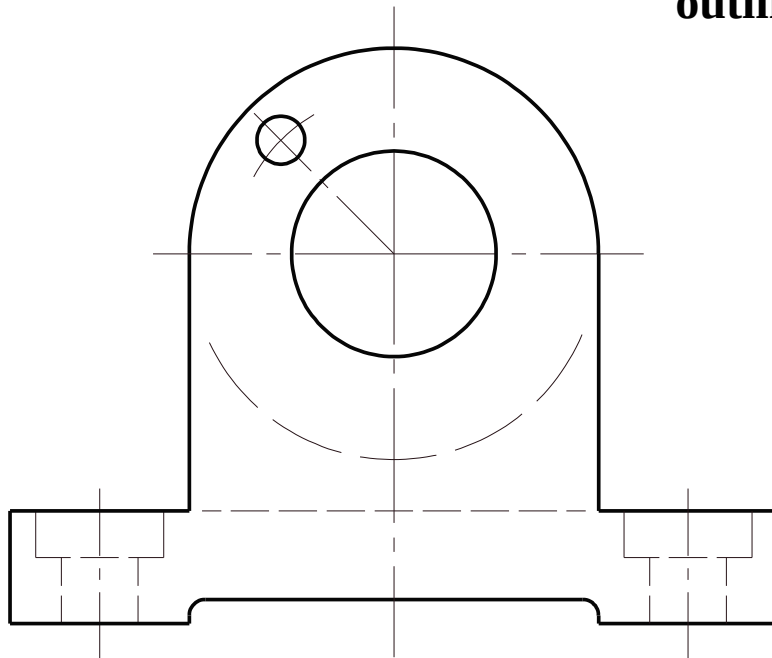
Long drawings should intersect

Out of profile

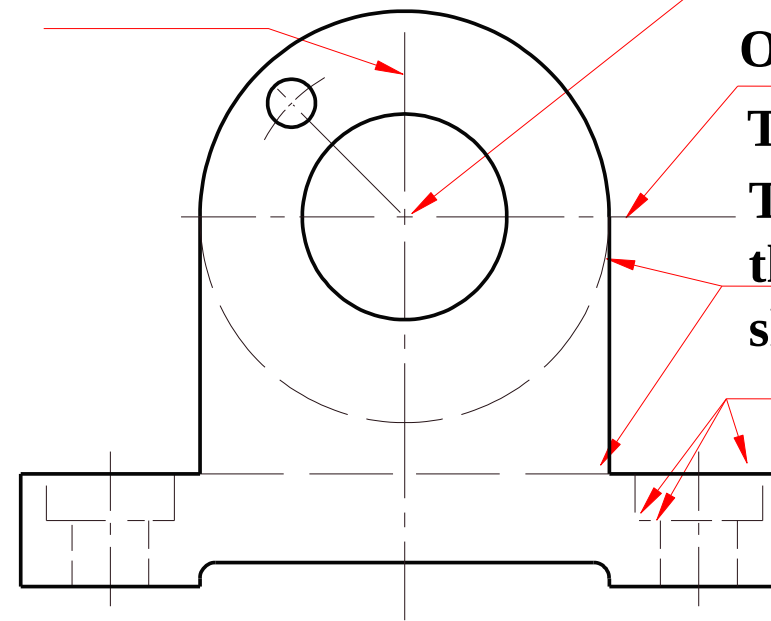
Thread too long

The extension of the thick line should be open

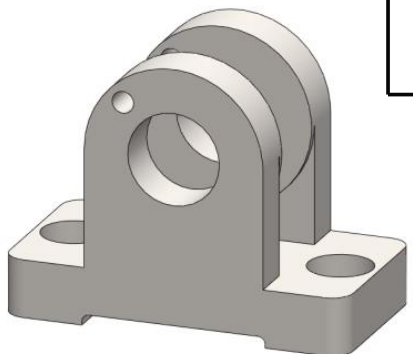
Segment intersection



True

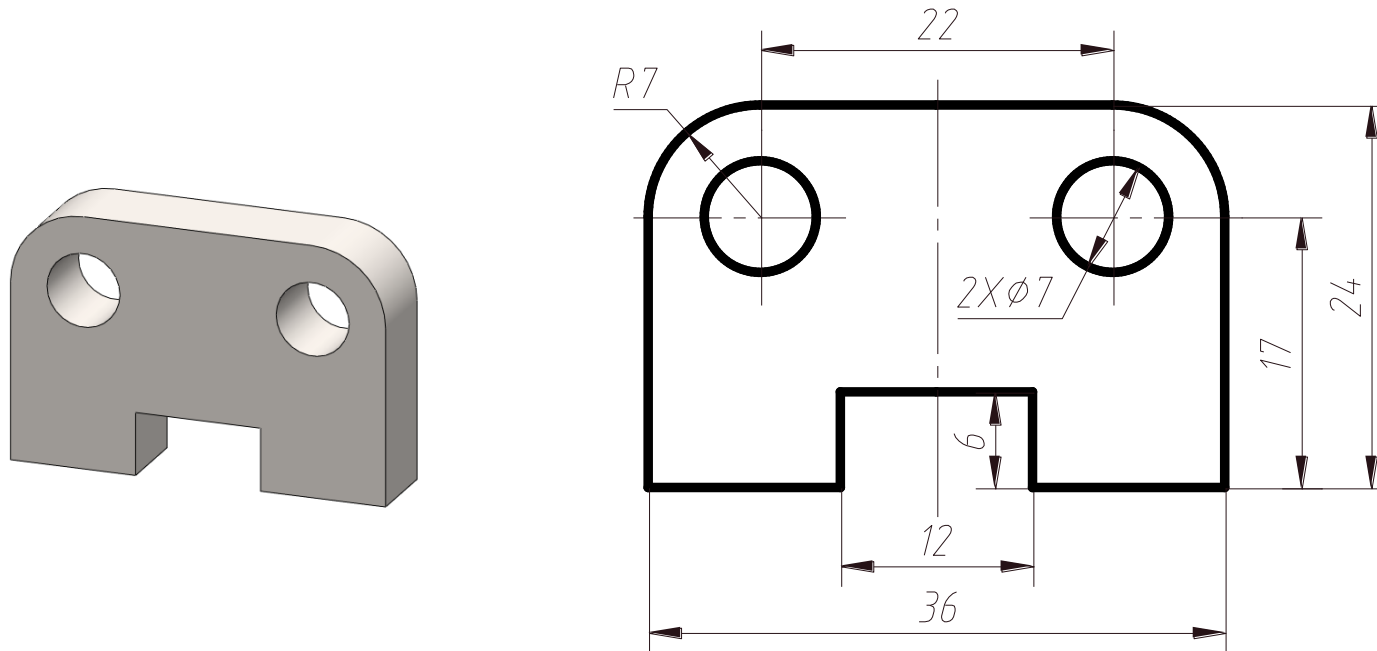


False



2、Drawing - dimensioning (GB/T 4458.4-2003)

The figure on the drawing can only represent the structural shape of the machine parts, and the size of the machine parts is based on the size value on the drawing for manufacturing and inspection.

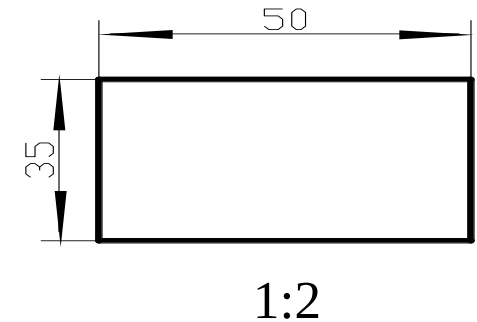
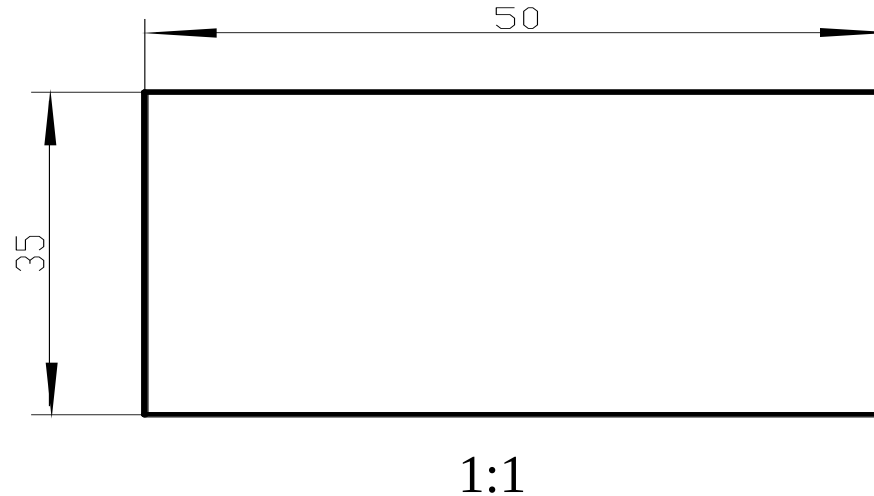


The national standard stipulates the rules and methods for marking the size, which must be strictly observed when drawing, otherwise it will cause confusion and bring difficulties and losses to production.

2、 Drawing - dimensioning (GB/T 4458.4-2003)

●The basic rules:

(1) True size



2、 Drawing - dimensioning (GB/T 4458.4-2003)

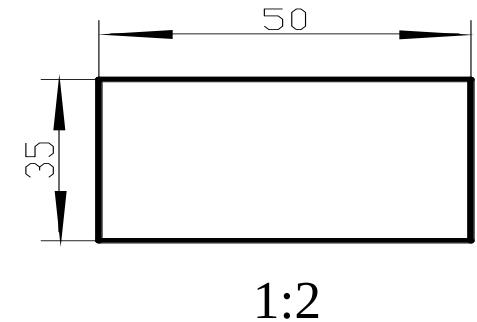
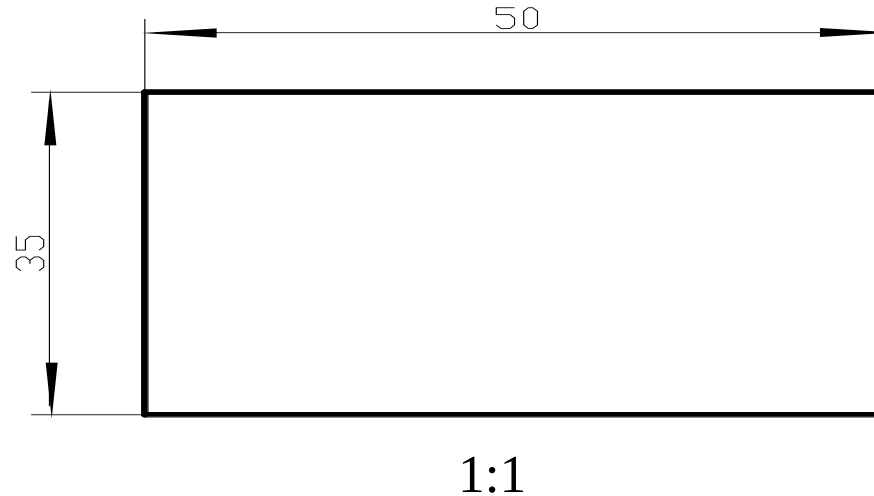
●The basic rules:

(1) True size

(2) In **millimetres**

(3) Finished dimensions

(4) Each size is marked only once



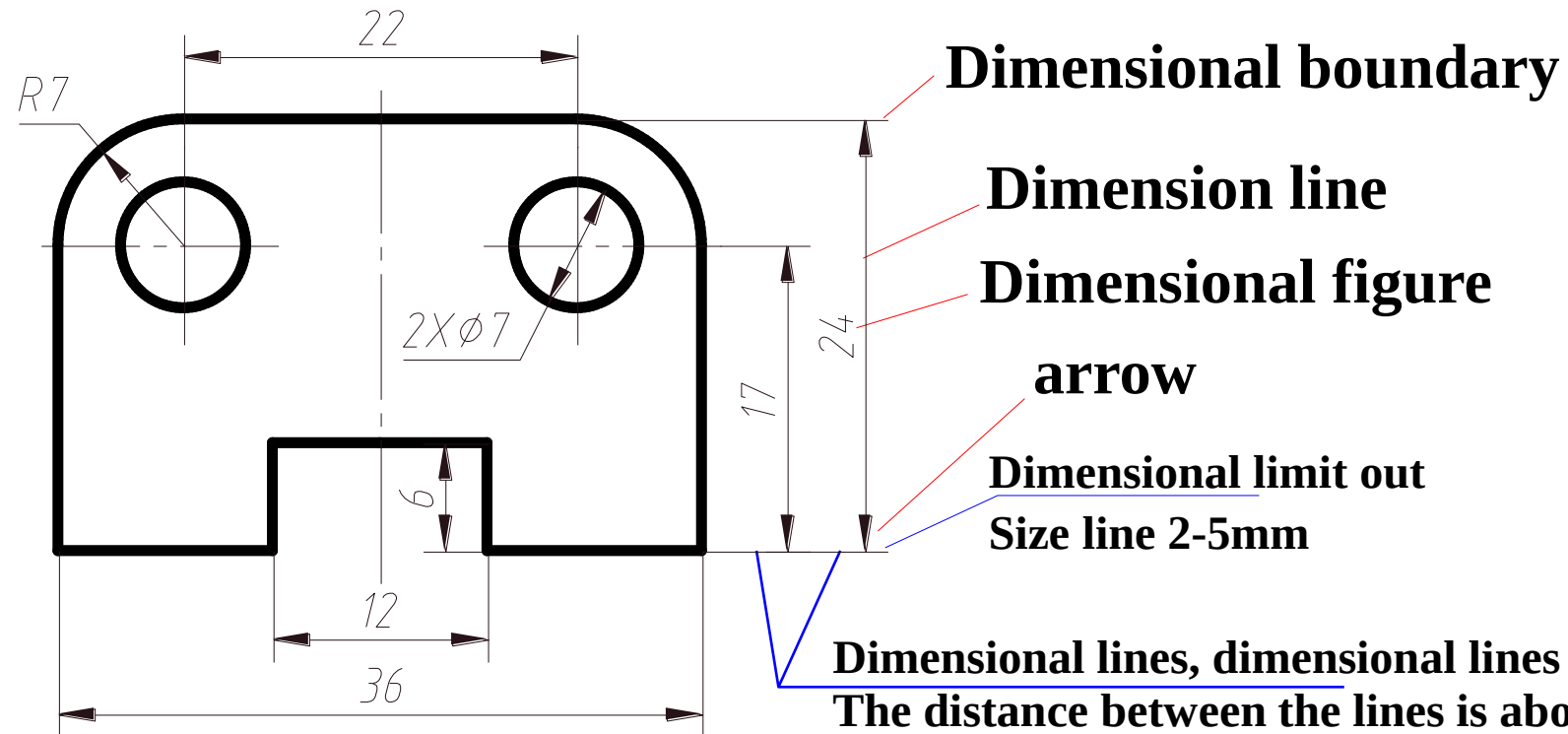
Note the unit symbol not in mm



● Dimensional composition

Four dimensions

- (1) Size boundary (thin solid line)
- (2) Dimension line (thin solid line)
- (3) Size number (text)
- (4) Dimension terminal (arrow)



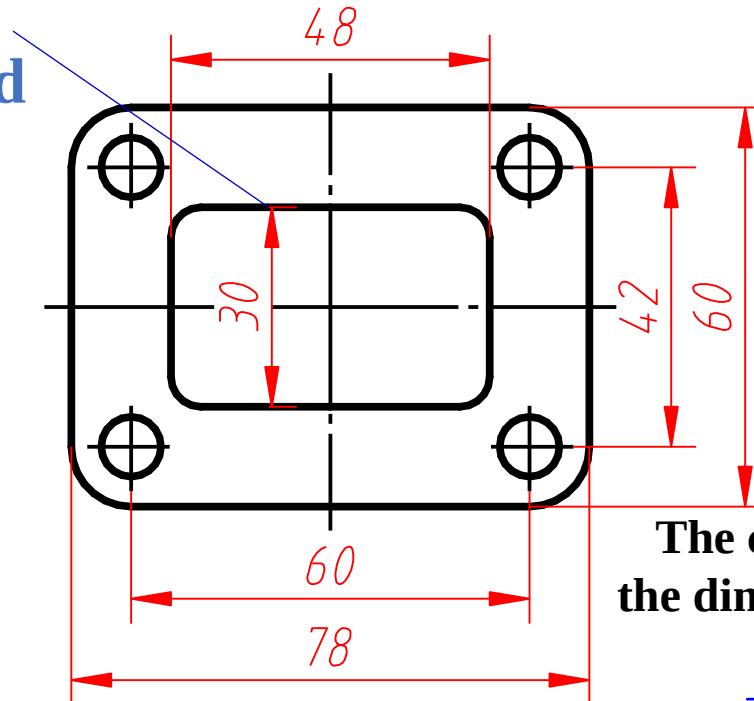
Dimensional lines, dimension lines and Outlines
The distance between the lines is about
2 times the height of the word,
And big size on the outside, small size on the inside

● Dimensional boundary

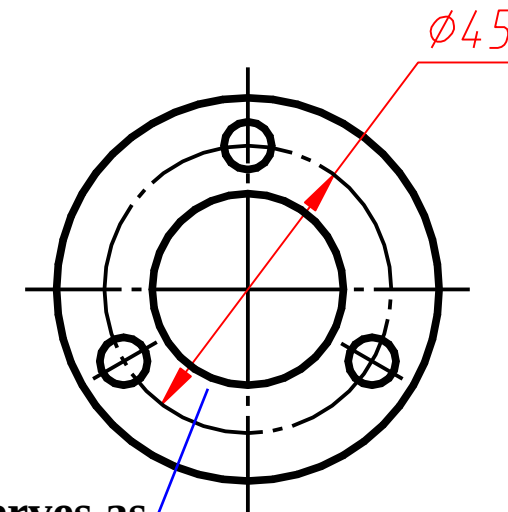
Position: Contour line, axis, symmetrical center line, and extension line,
also can use contour line

Line shape: thin solid line

Contour lines are used
as dimensional
boundaries



The center line serves as
the dimensional boundary



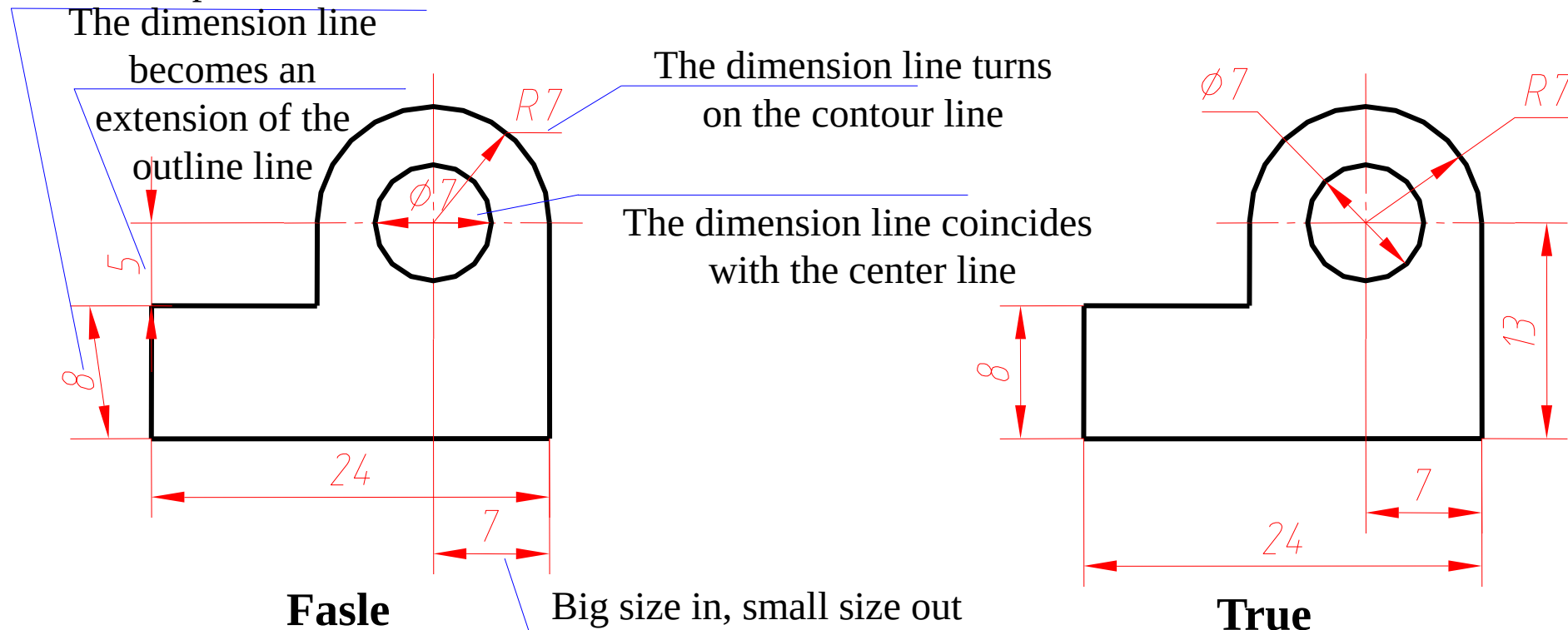
● Dimension line

Features: **Must be drawn separately**; It is not allowed to borrow other drawing lines or extension lines of other drawing lines

Position: For linear dimensions, **parallel to the marked line segment**; Generally, it should be perpendicular to the size line

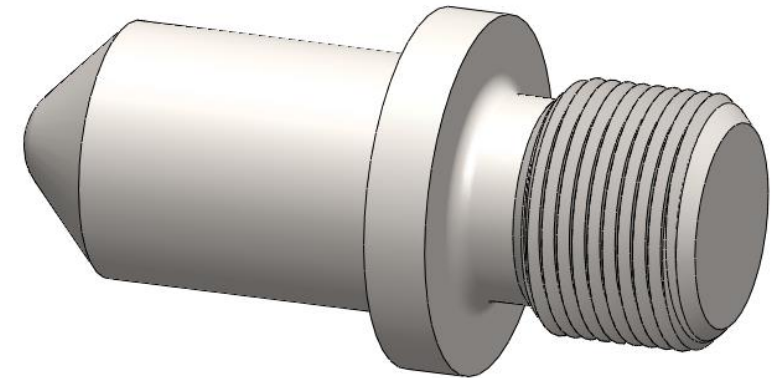
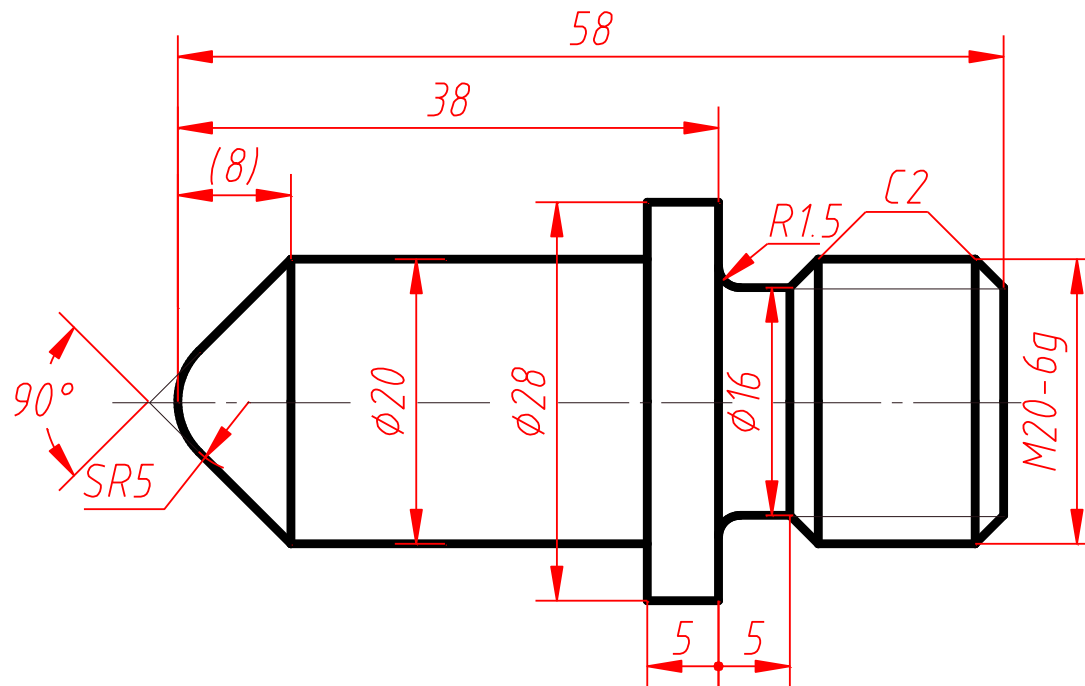
Line shape: thin solid line

The dimension line is not parallel to the outline line

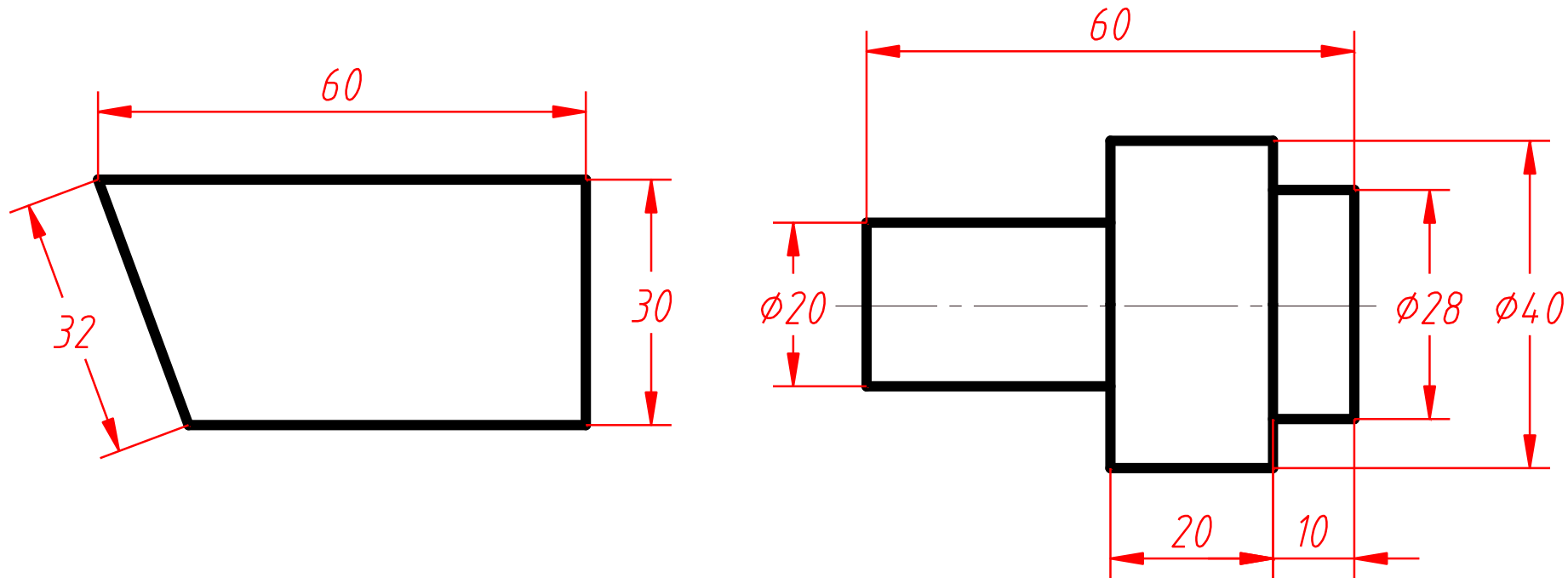


● Dimensional figure

1) The **linear dimension horizontal direction** of the number should generally be **written above the size line**, the vertical direction of the size of the number **written on the left side** of the size line, the head of the word to the left. It is also allowed to write on the center of the dimension line.

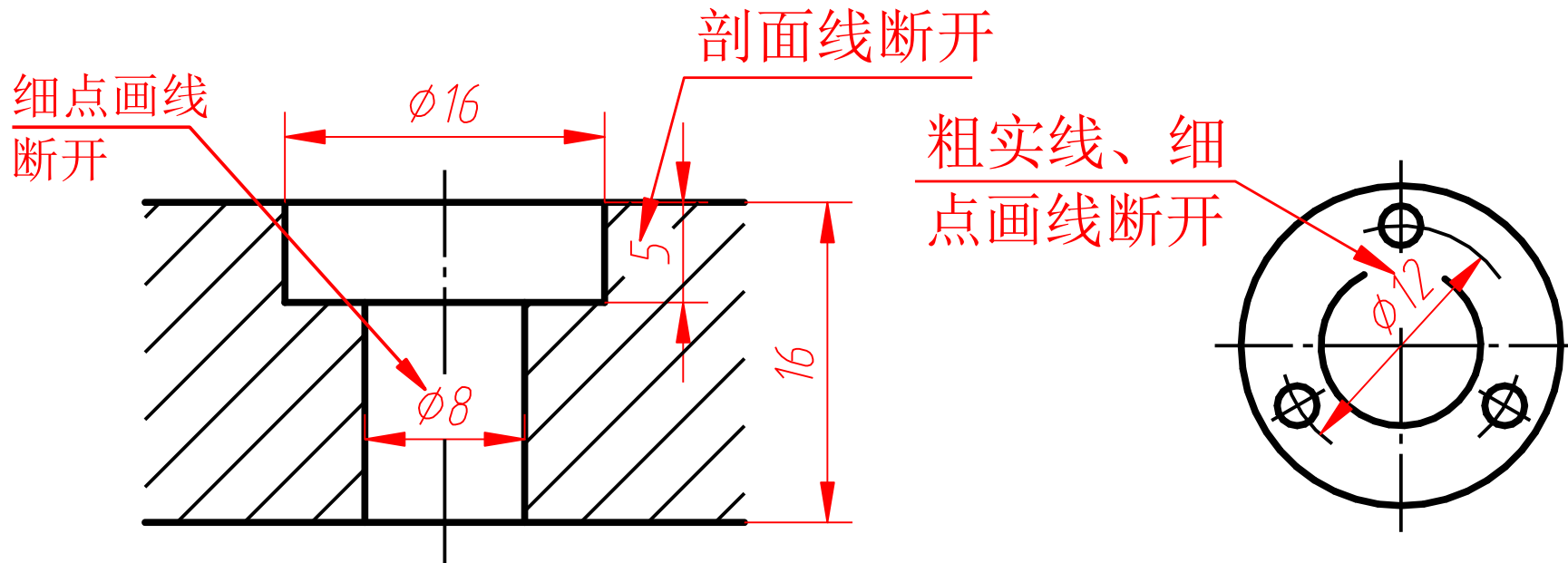


2) When marking the **reference size**, the size number should be in **parentheses**, such as the number (8)



- 3) **For non-horizontal dimensions**, the number can be written horizontally at the center of the dimension line.
- 4) **In the same drawing**, it should be marked **in the same form as far as possible**, and the size of the number is consistent.

- 5) The size number is not allowed to be passed by any drawing line, otherwise the drawing line **must be disconnected**.



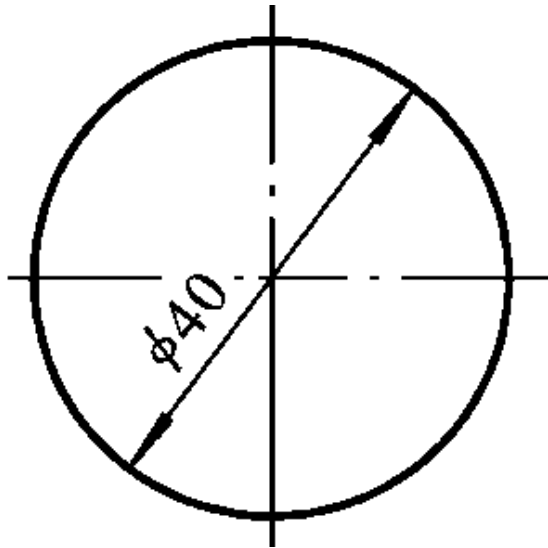
● Size symbols and abbreviations

尺寸的相关符号及缩写词

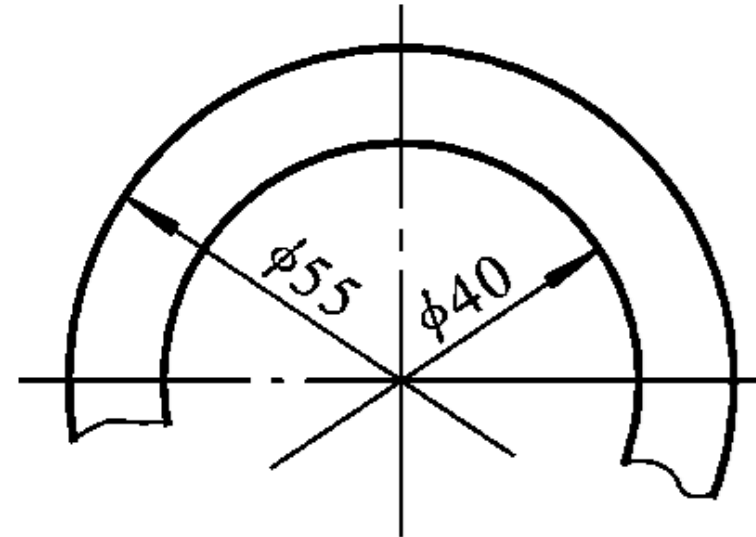
符 号	含 义	符 号	含 义
ϕ	直径	t	厚度
R	半径	$\sphericalangle$	埋头孔
$S\phi(R)$	球直径(半径)	$\sqcup$	沉孔
EQS	均布	$\downarrow$	深度
C	45°倒角	$\square$	正方形
$\angle$	斜度	∇	锥度

● Circular arc class annotation

Whole circle



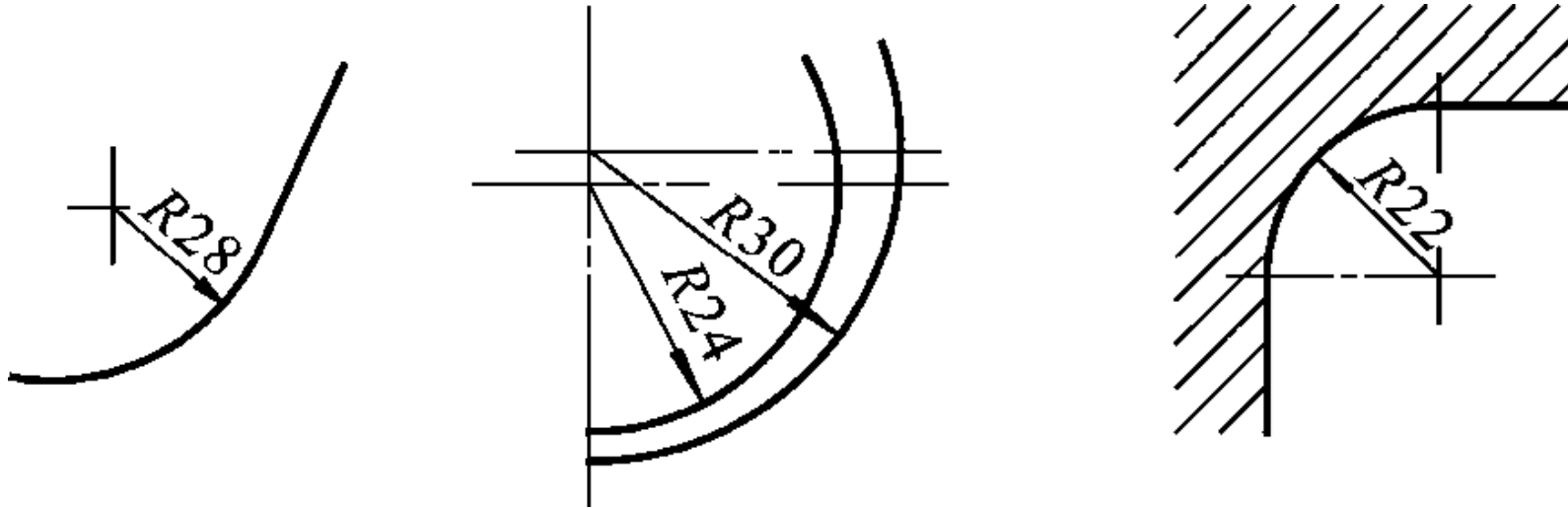
Greater than semicircle



**Must pass through the center of the circle
and mark the diameter**

● Circular arc class annotation

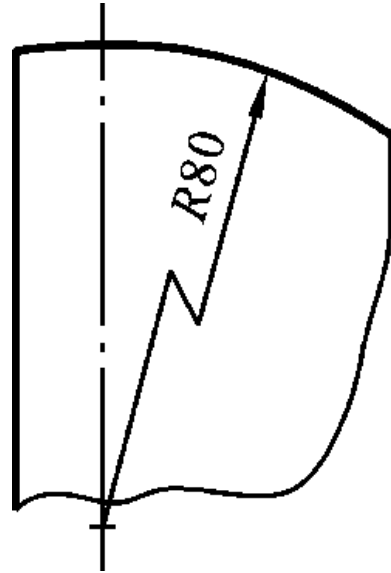
Less than or equal to a semicircle



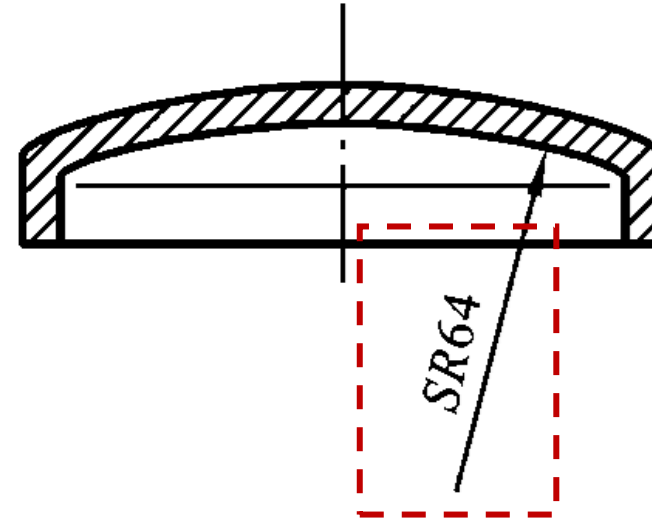
It must pass through the center of the circle and mark the radius

● Circular arc class annotation

Large circular radius

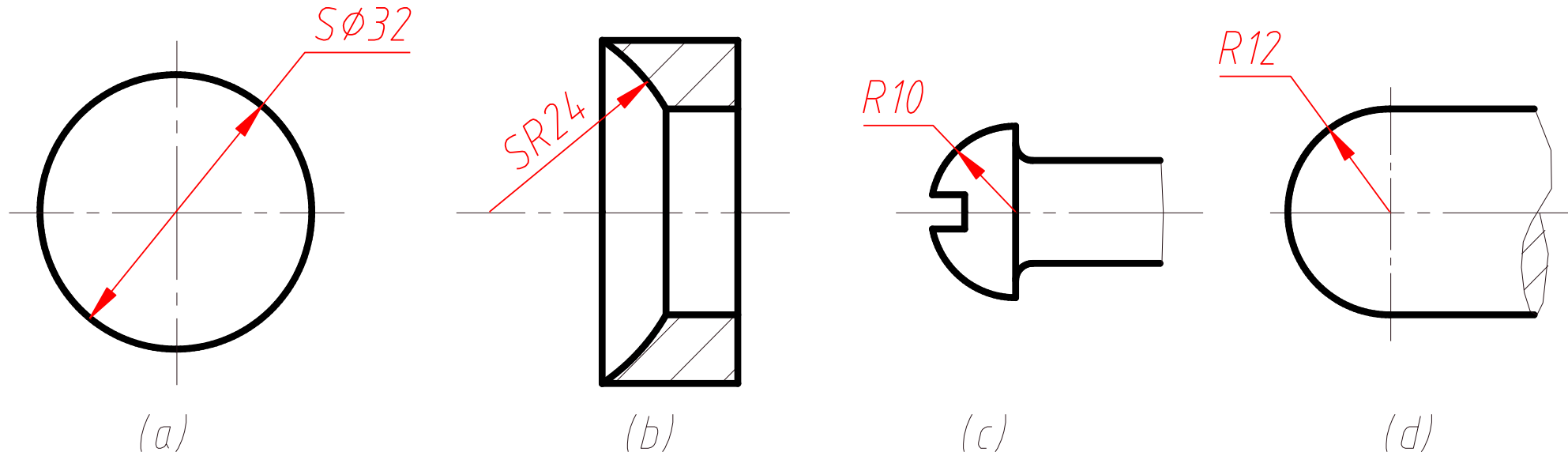


Point out that the center of the circle is on a symmetric line



The direction of the dimension line points to the center of the circle

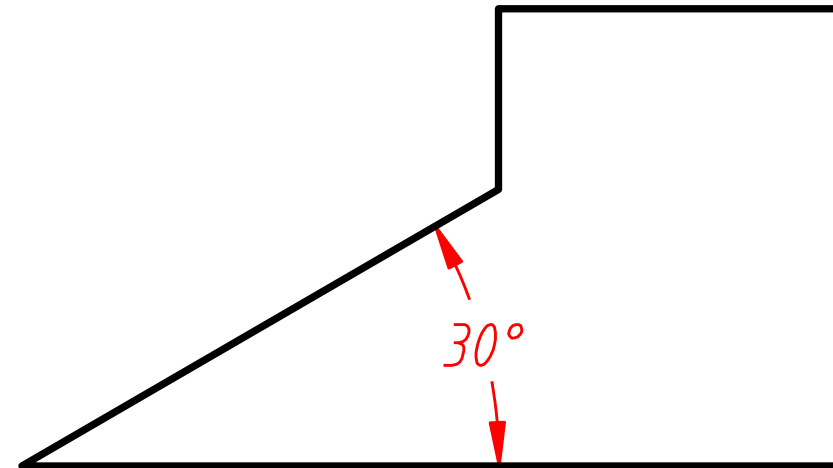
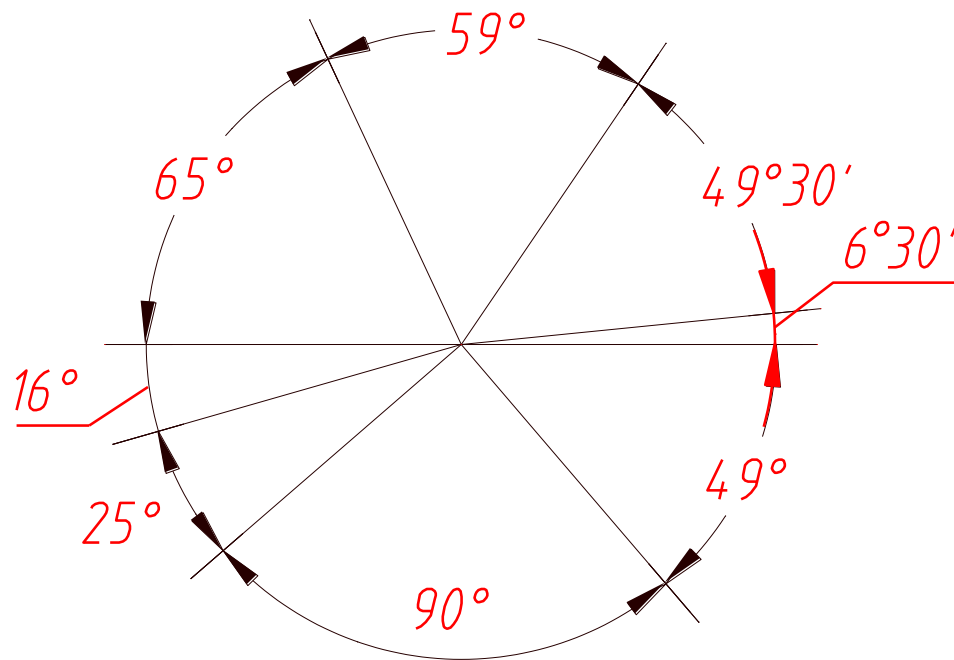
● Circular arc class annotation



- When marking the diameter or radius of the sphere, S diameter or SR should be applied (Figures a and b).
- For the head of the screw, the head of the rivet, the end of the shaft (including the screw) and the end of the handle, etc., the symbol S can be omitted if it does not cause misunderstanding (figures c, d).

● Angle labeling

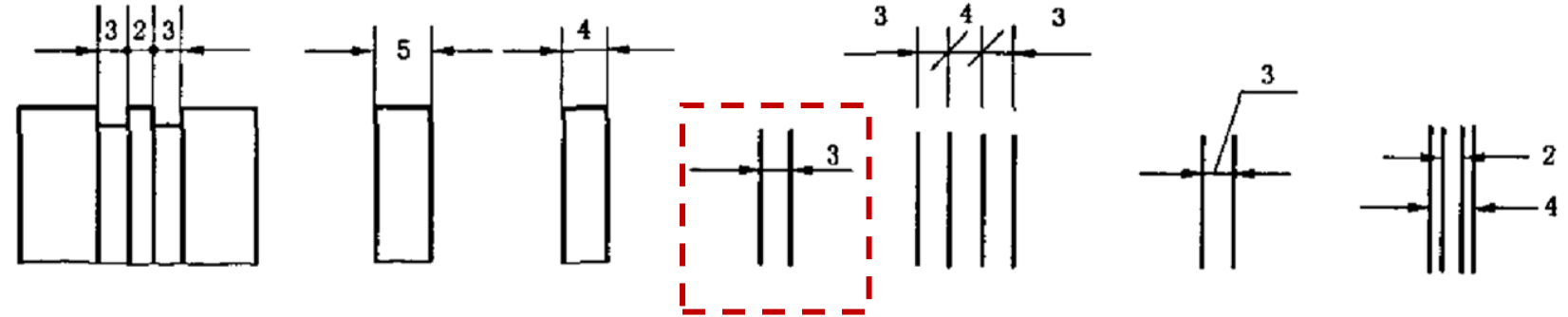
Angle number horizontal direction



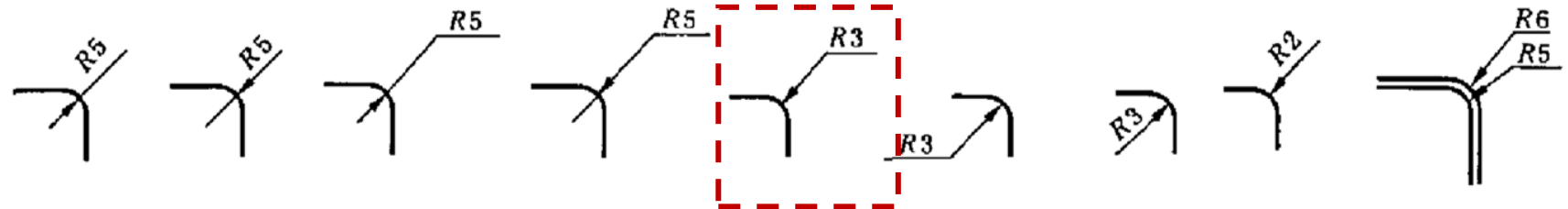
Usually the note is written at the center of the dimension line

● Narrow location annotation

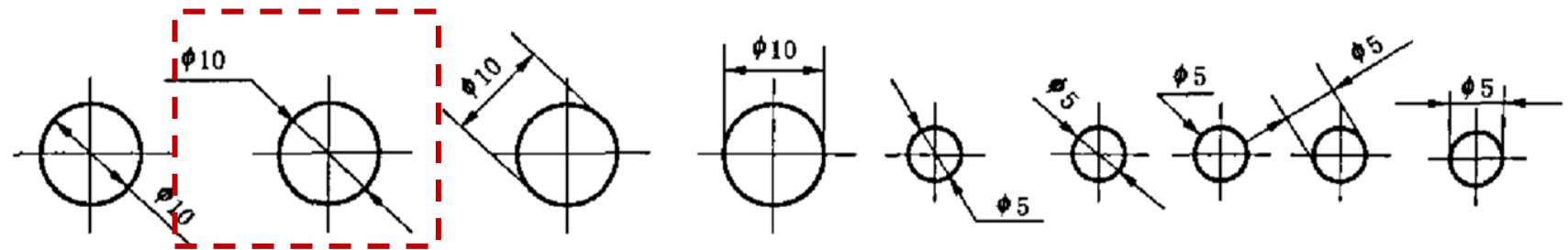
Straight line segment:
Dots, slashes, leads



Small arc:
Toward the center of the circle



Small circle:
Over the center of the circle



2、 Pattern - Typeface standard (GB/T 14691-1993)

Typeface refers to the writing form of **words, letters, and numbers** in the drawing.

1. 基本要求

(1) 书写字体必须做到：字体工整、笔画清楚、间隔均匀、排列整齐。

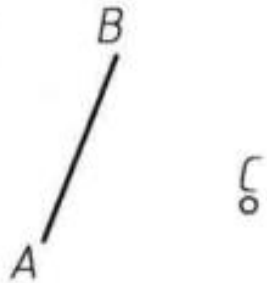
(2) 字体高度(用 h 表示)必须规范,其公称尺寸系列为: 1.8mm, 2.5mm, 3.5mm, 5mm, 7mm, 10mm, 14mm, 20mm。如需书写更大的字,其高度应按 $\sqrt{2}$ 的比率递增。不同高度的字体以不同号数称之,字体高度代表字体的号数,如 5mm 高的字体称为 5 号字。

(3) 汉字应写成长仿宋体字,并应采用中华人民共和国国务院正式公布推行的《汉字简化方案》中规定的简化字。汉字的高度 h 不应小于 3.5mm,其字宽一般为 $h/\sqrt{2}$ (约 $0.7h$)。汉字示例如图 2-7 所示。书写的要点在于横平竖直,注意起落,结构均匀,填满方格。

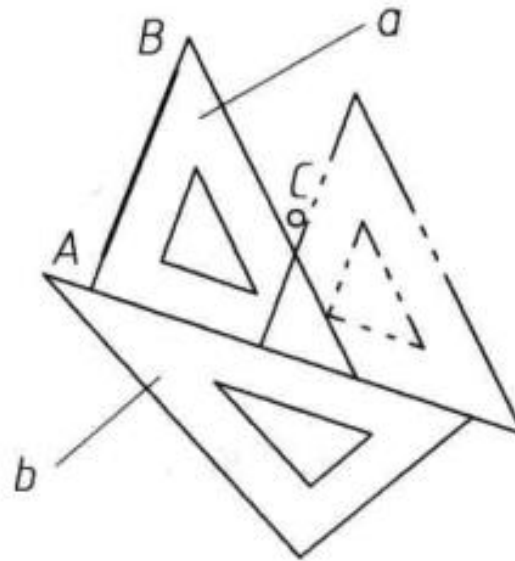
- **Projection Method**
- **National Standard**
- **Scale Drawing**

1、 Parallel lines

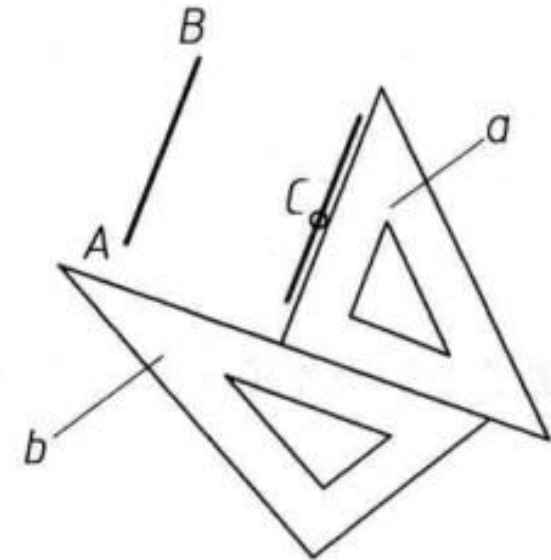
Make parallel lines through a given point



(1) 已知直线AB和点C



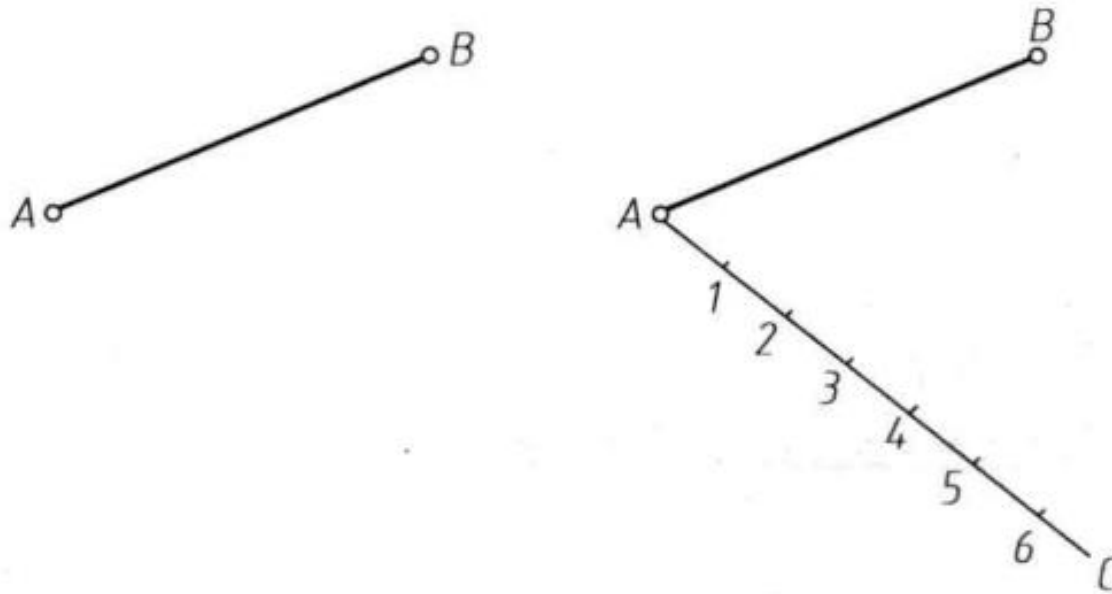
(2) 令三角板a一直角边与AB重合，三角板b一边与a另一直角边紧贴



(3) 按住b不动，令a沿b的一边滑动，当a的直角边与点C重合时，作直线即可

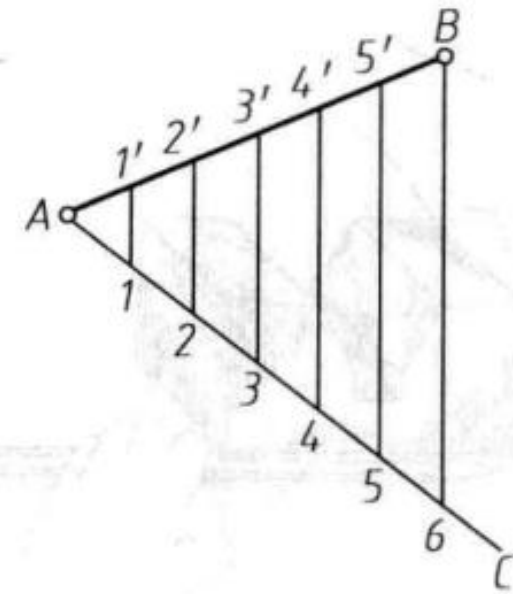
2、Line segments are arbitrarily divided

The dividing line segment is any equal division



(1) 已知直线AB

(2) 过线段的端点A(或B)任作一直线如AC; 自点A起在直线AC上, 以任意长度为单位线X取6个等分点, 得1, 2, 3, 4, 5, 6, 将线段AB分为六等分



(3) 连B6, 过AC上各等分点作B6的平行线与AB相交, 得1', 2', 3', 4', 5', 交点即为所求等分点

3、Regular polygon

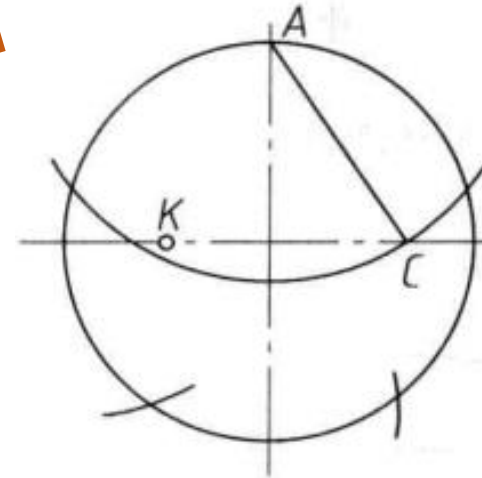
Known radius
of circle

Length of polygon side
= coefficient * radius
of the outer circle

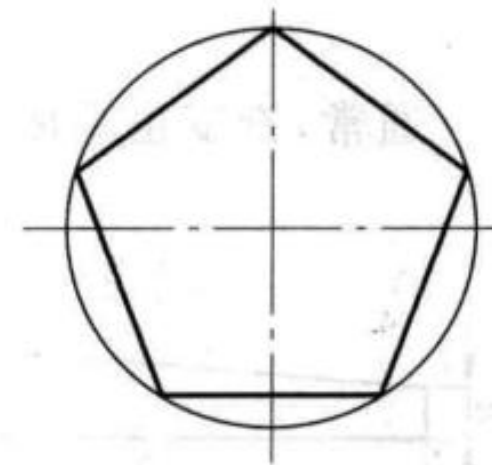
边长系数表

边数	系数	边数	系数
3	1.732	8	0.765
4	1.414	9	0.684
5	1.176	10	0.618
6	1.000	11	0.563
7	0.868	12	0.518

Enclosed pentagon



(3) AC即为五边形边长，
等分圆周得5个顶点

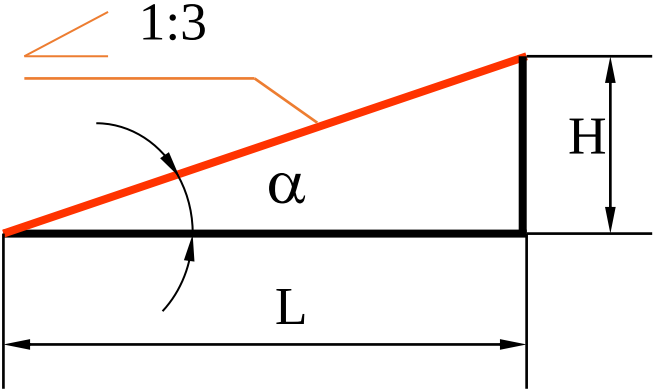
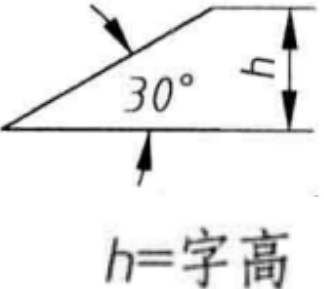
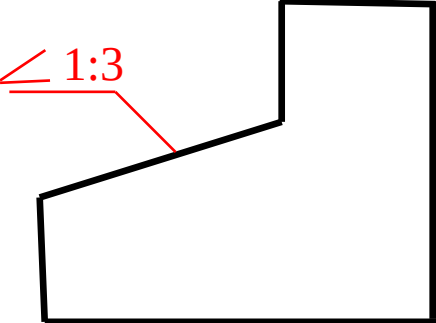
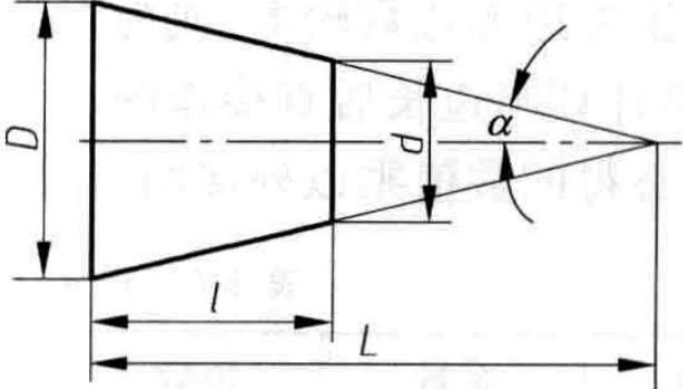
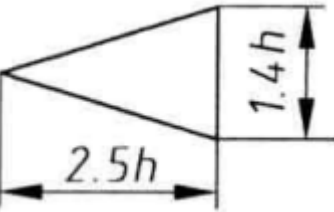
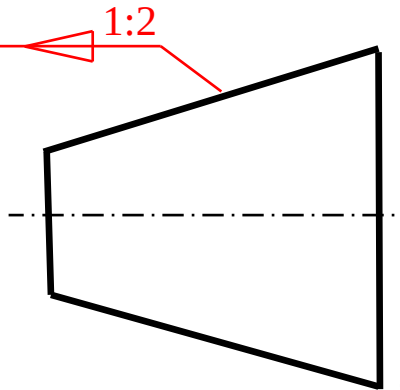


(4) 将5个顶点连线，
即成五边形

Given the length of
the sides of the polygon

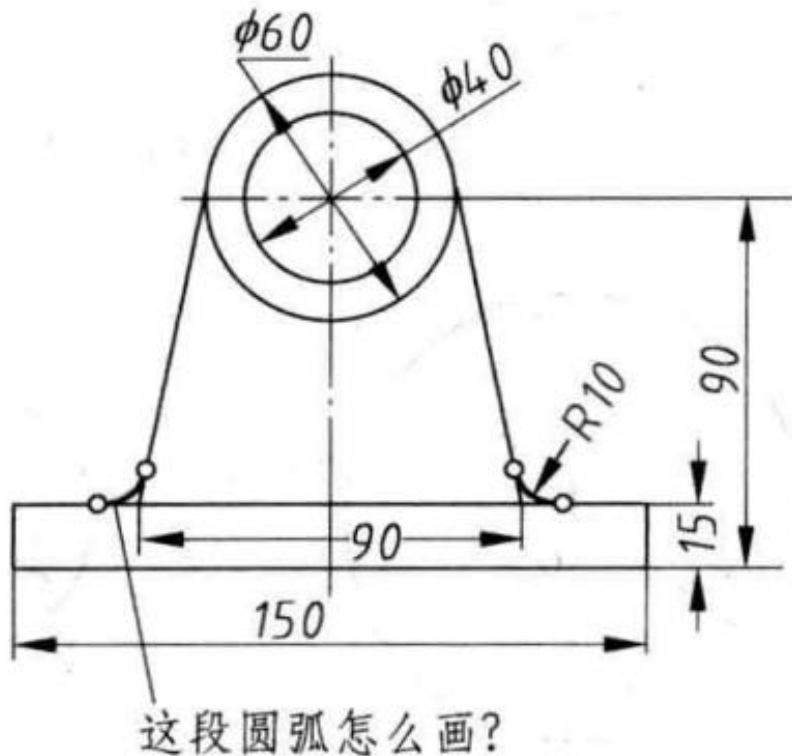
Radius of the outer circle =
Polygon side length/coefficient

4、Slope and taper

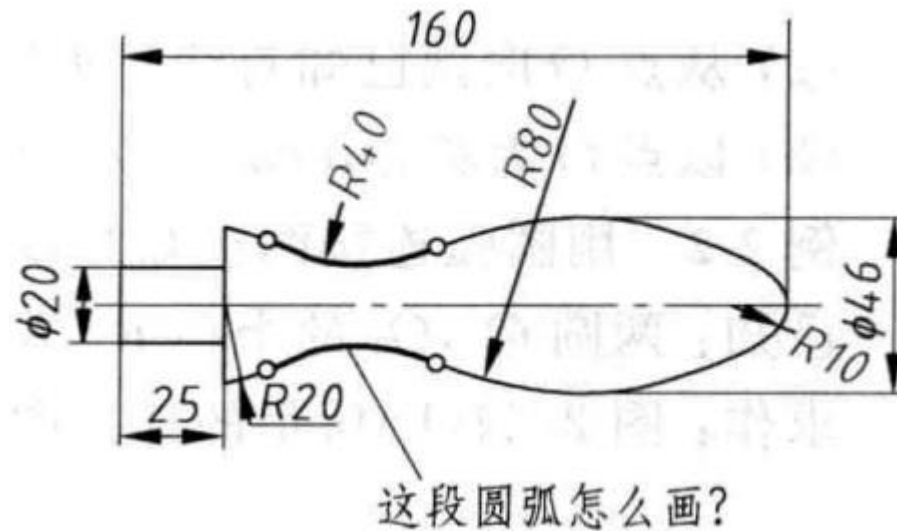
	Definition	Symbol	Calculation formula	Example
Slope			$\text{斜度} = \tan \alpha = H:L$	
Taper			$\begin{aligned} \text{锥度} &= 2 \tan \alpha \\ &= D:L = (D-d):l \end{aligned}$	

5、Arc connection - three tangential forms

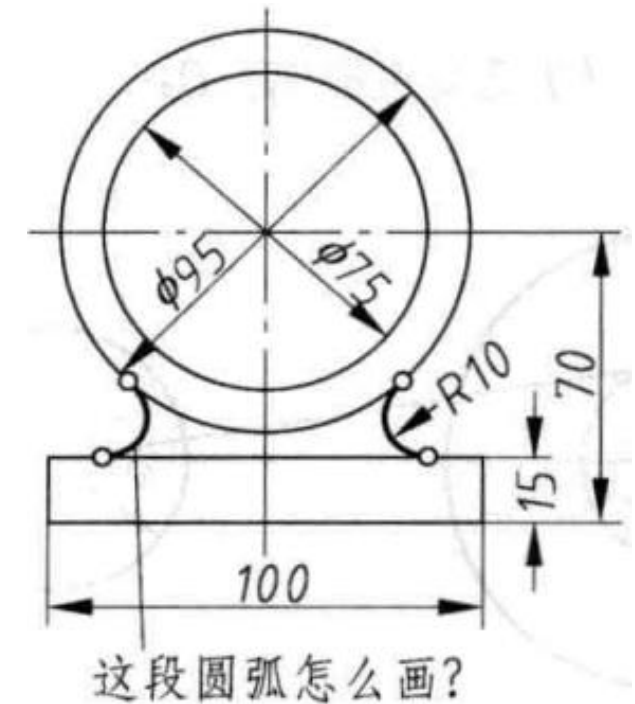
- Straight line - Arc -
Straight line



- Arc - Arc - Arc

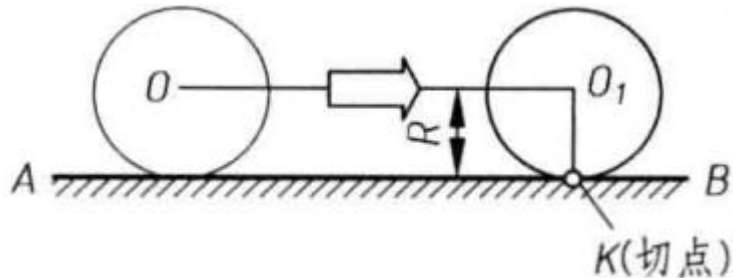


- Line - Arc - Arc

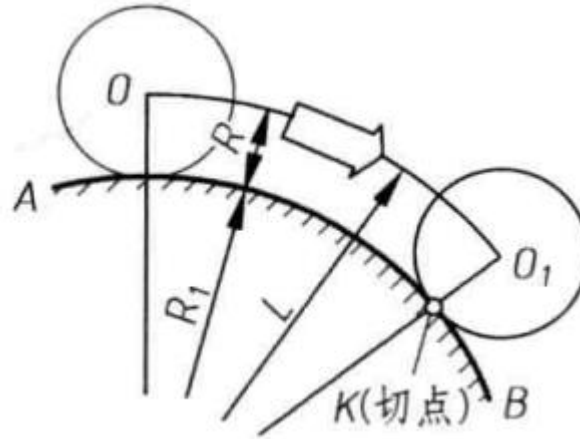


5、Arc connection-tangent principle

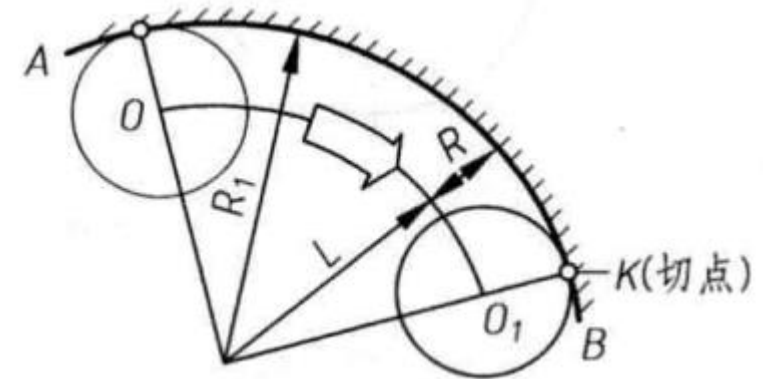
- The circle is tangent to the line



- Circle - The outer cut of the circle



- Circle - The inner cut of the circle



Center trajectory:
parallel to the line

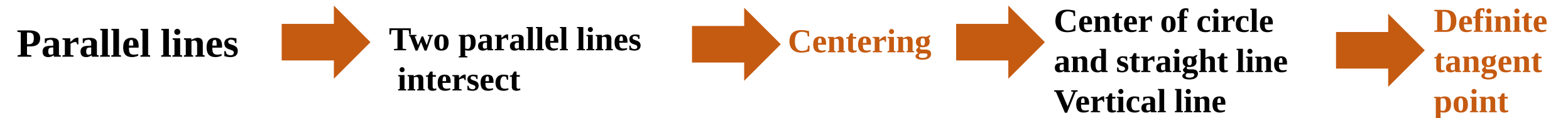
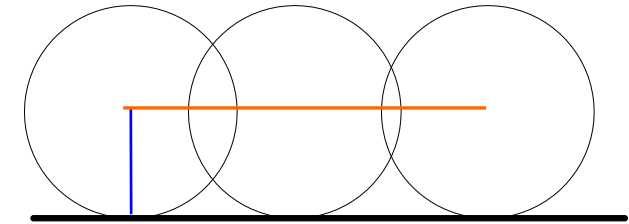
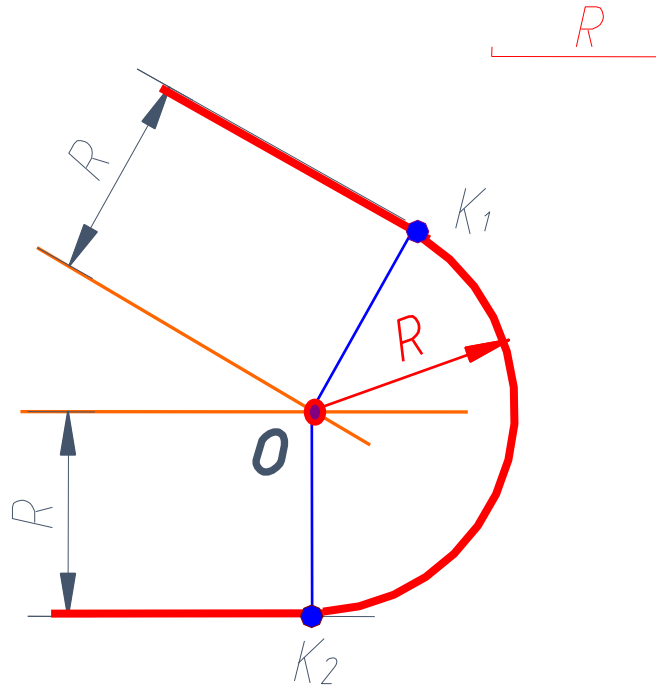
Tangential position:
The vertical intersection of two parallel lines

Center trajectory: the known center of the circle as the center of the circle, the radius of the superimposed arc

Cutting point position: the point at which the center of two circles is connected to the intersection of two circles

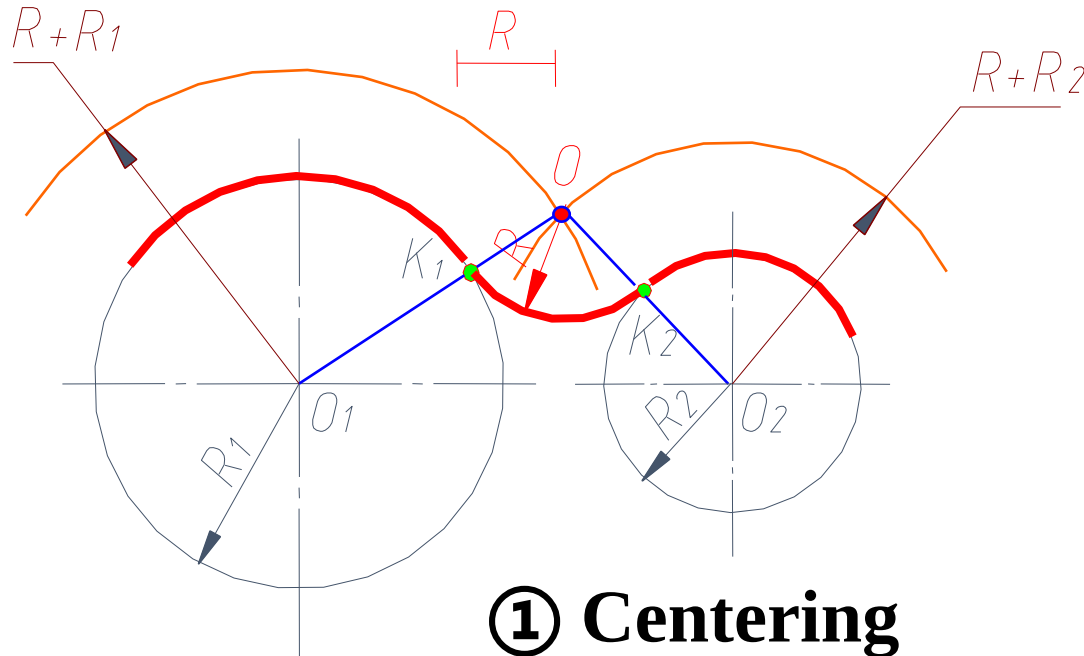
5、Arc connection -(line - arc - line)

- ① Centering
- ② Fixed point of contact
- ③ Arc



5、Arc connection -(arc - arc - arc)

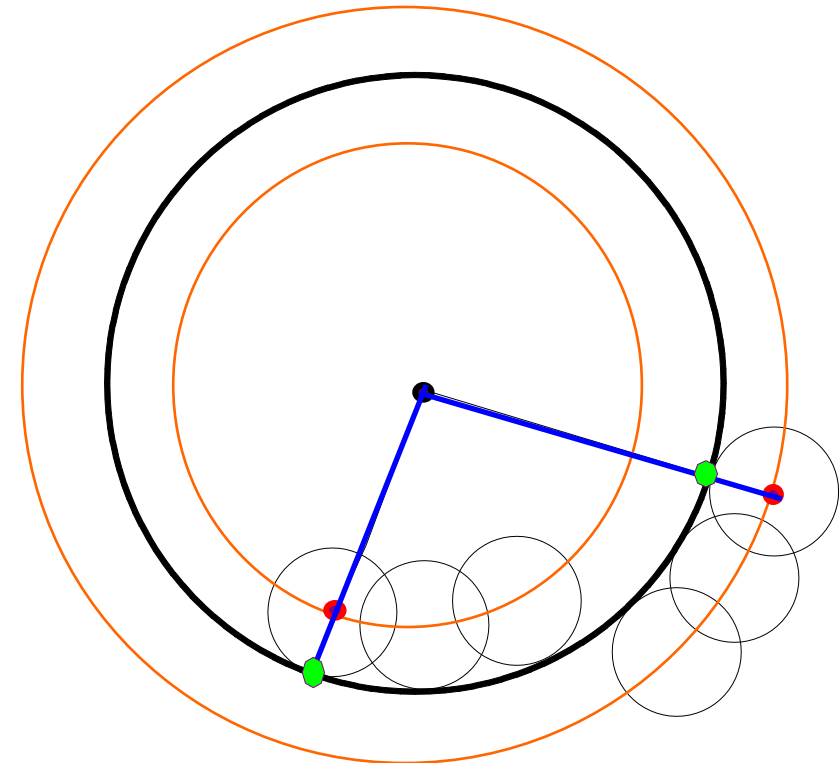
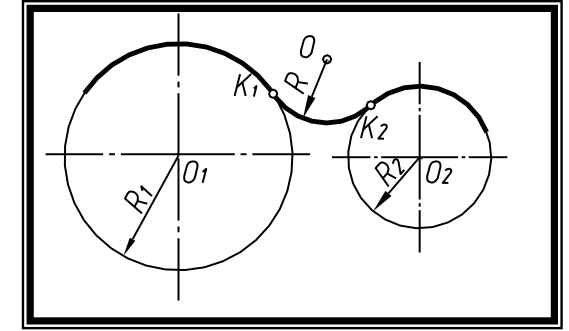
External arc



① Centering

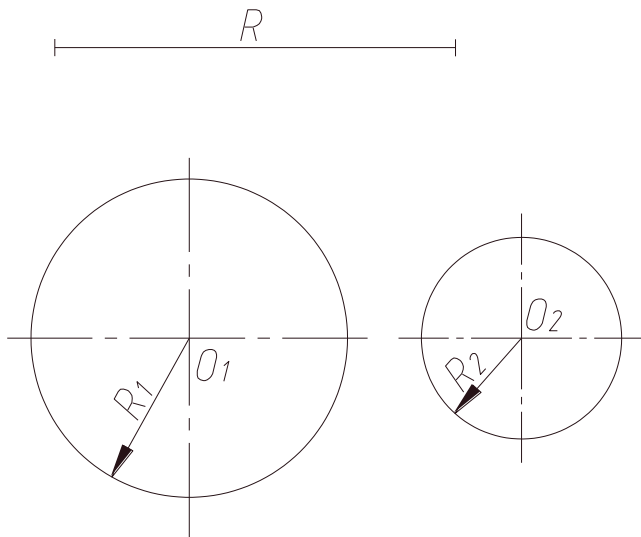
② Fixed point of contact

③ Arc

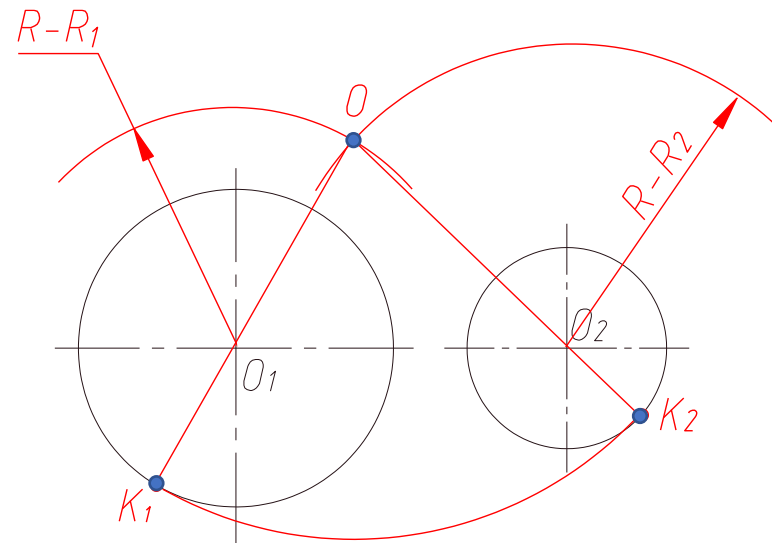


The tangent circle arc

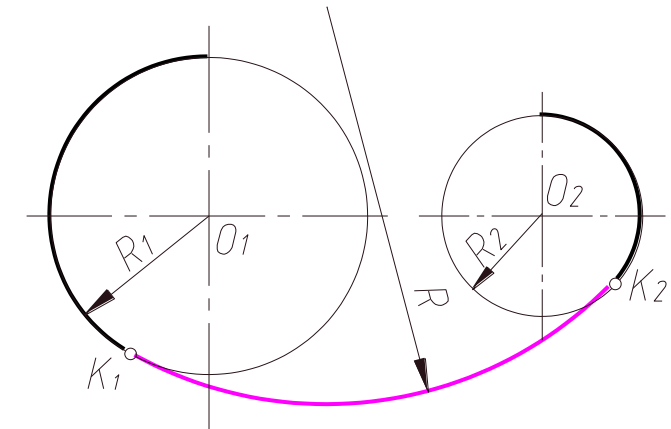
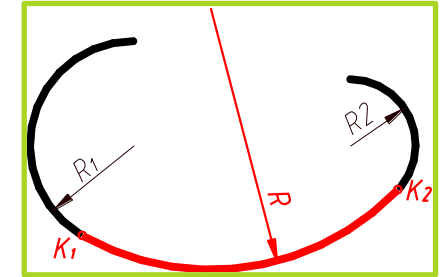
5. Arc connection -(arc - arc - arc)



a) Two arcs and the radius of the connecting arcs are known



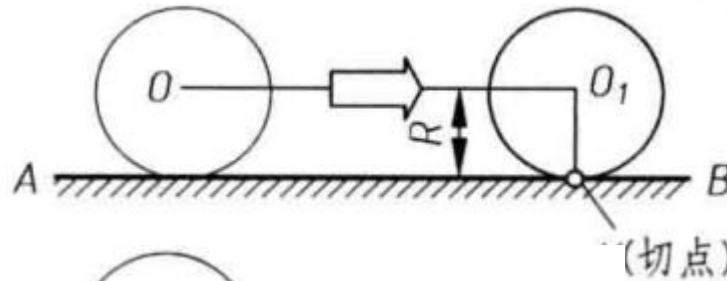
b) Find the center of the circle (minus the radius) of the connecting arc and its tangent point to the known arc



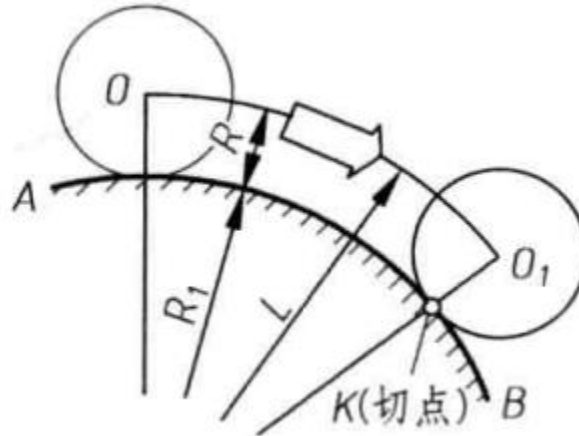
c) Draw the connecting arcs

5、Arc connection - Summary

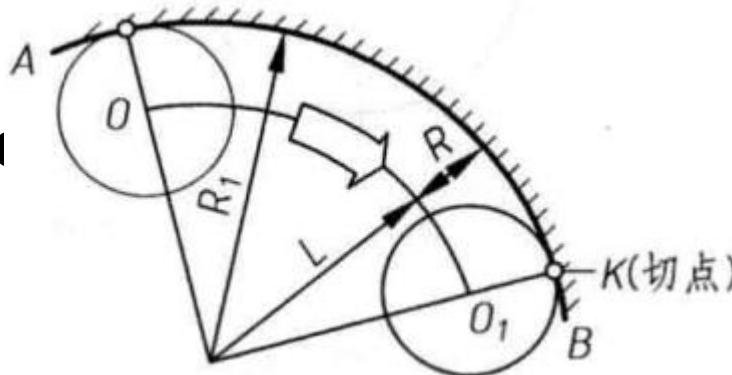
- The circle is tangent to the line



- Circle - The outer cut of the circle



- Circle - The inner cut of the circle



Sum up

Straight lines are tangent: make parallel lines

External tangent of an arc: an arc that adds radii

Arc tangent: An arc with a subtracting radius